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Central Bank Foreign Currency Swaps and Repo Facilities Survey

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Abstract

Central bank swap arrangements gained new importance during the Global Financial Crisis of 2007–09 (GFC) when wholesale funding markets contracted, causing severe liquidity strains. Led by the Federal Reserve, central banks, in their roles as lenders of last resort, redeployed this tool to provide liquidity in their currencies across borders to great effect. Since the GFC, swap arrangements have become key central bank policy tools and have been repeatedly used to address liquidity constraints. In this paper, we survey 67 swap and swap-like repurchase arrangements and frameworks from the 20th and 21st centuries. In addition, we analyze key design decisions of interest to policymakers seeking to design a central bank swap arrangement. We also review structural developments regarding this policy tool, such as the increasing use of repurchase (repo) arrangements in addition to swaps and multilateral regional swap facilities. We conclude that swap arrangements and related repo arrangements have proved an efficient response to liquidity strains, have become a standing tool to enhance the global financial stability safety net, and are likely to see continued use.

Keywords: central bank cooperation, central bank currency swaps, financial stability, lender of last resort, liquidity strain, repurchase agreement

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Introductory Note: This survey is an analysis of important considerations for policymakers seeking to establish a central bank currency swap arrangement or a similar repurchase (repo) agreement. It is based on insights derived from 23 recently completed Yale Program on Financial Stability case studies and the existing literature on the topic. While this survey can help inform a decision about whether or not to establish a swap or repo arrangement, our main purpose is to assist policymakers who have already made that decision in designing the most effective program possible. In analyzing the programs that are the focus of this survey, we used a color-coded system to highlight certain particularly noteworthy design features.

Treatment	Meaning
BLUE – INTERESTING	A design feature that is interesting and that policymakers may want to consider. Typically, this determination is based on the observation that the design feature involves a unique and potentially promising way of addressing a challenge common to this type of program that may not be obvious. Less commonly, empirical evidence or a consensus will indicate that the design feature was effective in this context, in which case we will describe that evidence or consensus.
YELLOW – CAUTION INDICATED	A design feature that policymakers should exercise caution in considering. Typically, this determination is based on the observation that the designers of the feature later made significant changes to the feature with the intention of improving the program. Less commonly, empirical evidence or a consensus will indicate that the design feature was ineffective in this context, in which case we will describe that evidence or consensus.
FOOTNOTE IN ITALICS	Where the reason that a given design feature has been highlighted is not apparent from the text, it is accompanied by an italicized footnote that explains why we chose to highlight it. Where necessary, these footnotes will be used to identify any considerations that should be kept in mind when thinking about the feature.

This highlighting is not intended to be dispositive. The fact that a design feature is not highlighted or is highlighted yellow does not mean that it should not be considered or that it will never be effective under any circumstances. Similarly, the fact that a design feature is not highlighted or is highlighted blue does not mean that it should always be considered or will be effective under all circumstances. The highlighting is our subjective attempt to guide readers toward certain design features that (1) may not be obvious but are worth considering or (2) require caution.

Introduction

Modern trade and financial systems are highly interconnected through global institutions and markets. The five dominant reserve currencies—US dollars, euros, yen, pounds sterling, and renminbi—figure in transactions all over the world. Distress in a currency's home country can spill over to foreign markets, causing destabilizing liquidity strains and threatening foreign institutions. Likewise, pressures that develop outside the country are at risk of migrating into the home country. In crises, banks become reluctant to lend to one another because of concerns about the financial condition of their potential counterparties (Steil, Della Rocca, and Walker 2021). Central banks, in their roles as lenders of last resort (LOLRs), with their mandates of maintaining financial stability, also must consider the cross-border implications of any liquidity stresses.

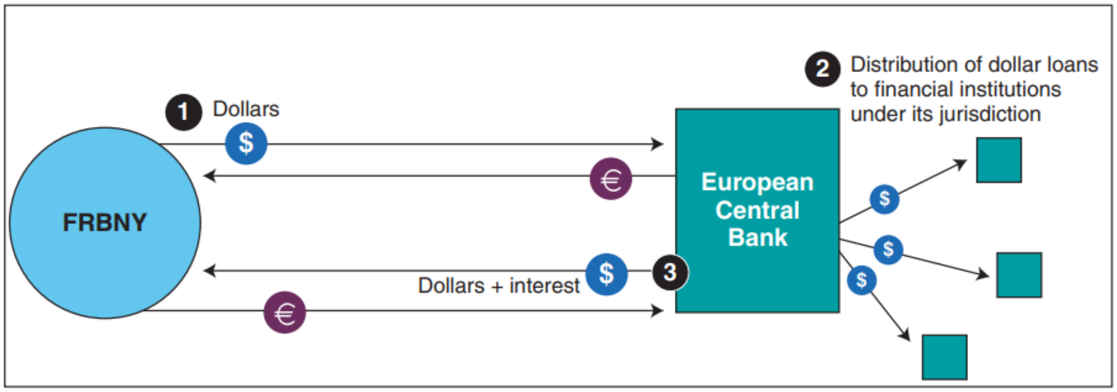
For these reasons, swap arrangements have become key tools of central bank cooperation since the Global Financial Crisis of 2007–09 (GFC) (McCauley and Schenk 2020). The use of swaps has been "hailed as one of the most important and effective policy responses to the financial crisis" (Awrey 2017, 935) and as the "main innovation in central bank cooperation" during the GFC (Allen and Moessner 2010, 1). Scholars and government officials have found swap and swap-like arrangements to be highly effective in restoring liquidity in funding markets during periods of stress (Awrey 2017; Fleming and Klagge 2010; Goldberg and Ravazzolo 2022; Steil, Della Rocca, and Walker 2021).

In its simplest form, a central bank currency swap arrangement is an agreement between two central banks in which the parties agree that one bank (the "Lender," or "lending central bank") will sell its currency to the other (the "Borrower," or "borrowing central bank") for a specified period in exchange for an equivalent amount of the Borrower's currency as collateral at an agreed exchange rate, usually the spot rate[5] (Bahaj and Reis 2018). The two central banks also agree that, at maturity, the Lender will repurchase its currency from the Borrower at a forward exchange rate equal to the original spot exchange rate and return the collateral to the Borrower. The Borrower will also pay to the Lender an agreed-upon amount of interest, based on a benchmark rate plus a premium; the Lender pays no interest (Bahaj and Reis 2018).

Most swap agreements specify a maximum amount that may be borrowed and a maximum duration for draws (see Key Design Decision No. 9, Process for Utilizing the Swap Agreement). Swap agreements typically allow for repeated drawdowns, similar to a master agreement for a revolving line of credit. At the time of a draw, the parties agree on certain transactional details, such as the amount and duration of the draw, the exchange rate, and the interest rate. The Borrower typically lends the drawn funds downstream to banks in its jurisdiction, denominated in the Lender's currency, as shown in Figure 1.

[5] The collateral is usually held by the Borrower in an account established for the benefit of the Lender, with the collateral remaining unused until the reversal of the swap.

Figure 1: A Typical Swap Transaction



Source: US GAO 2011.

This survey covers 23 cases (see Figure 2 for a complete list of cases) relating to 67 swap and swap-like repurchase (repo) arrangements and frameworks (see Figure 5 for a complete list of arrangements and frameworks)[6] established by central banks and international organizations. We include arrangements involving central banks of advanced economies (Group of 10 [G-10] countries) and emerging market economies. We cover 61 traditional swap lines (including three lines under the South Asian Association for Regional Cooperation [SAARC], 2012, framework), and two multilateral swap frameworks (Association of Southeast Asian Nations: ASEAN Swap Arrangement [ASA], 1977, and Association of Southeast Asian Nations Plus Three: The Chiang Mai Initiative [ASEAN +3 CMIM], 2010). Two cases relate to bilateral repo lines extended by the European Central Bank (ECB) to the central banks of Hungary and Poland. We also include cases about the ECB's Eurosystem repo facility for central banks (EUREP) and the Federal Reserve's Foreign and International Monetary Authorities (FIMA) Repo Facility, new broad-based facilities that were deployed in connection with the COVID-19 pandemic. See Key Design Decision No. 2, Purpose and Type (B), for further discussion.

In keeping with our mission and focus on crisis fighting, we have selected swap arrangements that were (1) originated to address an existing or eminent crisis or liquidity strain and (2) originated for other reasons, or not during a crisis, but were used to address a crisis. With the exception of the three People's Bank of China (PBOC) cases, we generally

[6] In writing the cases examined in this survey, we were able to obtain only a limited number of the swap agreements. Because of this restriction, the actual terms and legal structures of the swap arrangements are in many cases still not entirely clear. Additionally, many arrangements referred to by government officials, scholars, the media, and other parties as "swap agreements" or "swap lines" may have comprised more than one legal agreement. For these reasons, in this survey, we generally refer to the "swap arrangement" between parties and only refer to a "swap agreement" when referencing an actual agreement, which in many cases may have represented only one part of the arrangement. When an agreement was not available, we gleaned essential details of the swap arrangement from credible secondary sources.

chose not to include arrangements that the parties entered into strictly for currency stabilization or trade purposes.

Figure 2: Case Studies and Abbreviated Names

Case Name	Short-form Case Name	Reference
Association of Southeast Asian Nations: ASEAN Swap Arrangement, 1977	ASA 1977	Hoffner 2023a
Association of Southeast Asian Nations Plus Three: The Chiang Mai Initiative Multilateralization, 2010	ASEAN +3 CMIM 2010	Hoffner 2023b
China: Central Bank Swaps to Argentina, 2014	China–Argentina 2014	Arnold 2023b
China: Central Bank Swap to Hong Kong, 2009	China–Hong Kong 2009	Arnold 2023a
China: Central Bank Swaps to Mongolia, 2011	China–Mongolia 2011	Arnold 2023c
Eurozone: Central Bank Swap to Denmark, 2008	Eurozone–Denmark 2008	Gupta 2023c
Eurozone: EUREP, 2020	Eurozone–EUREP 2020	Arnold 2023d
Eurozone: Central Bank Repo to Hungary, 2008	Eurozone–Hungary Repo 2008	Gupta 2023a
Eurozone: Central Bank Swaps to Ireland and the United Kingdom, 2010	Eurozone–Ireland and UK 2010	Wiggins 2023
Eurozone: Central Bank Repo to Poland, 2008	Eurozone–Poland Repo 2008	Gupta 2023b
Eurozone: Central Bank Swap to Sweden, 2007	Eurozone–Sweden 2007	Gupta 2023d
Eurozone: Central Bank Swap to United Kingdom, 2019	Eurozone–UK 2019	Swaminathan 2023a
South Asian Association for Regional Cooperation: SAARC Swap Arrangement, 2012	SAARC 2012	Gupta 2023e
Scandinavia: Central Bank Swaps to Iceland, 2008	Scandinavia–Iceland 2008	Hoffner 2023c
Switzerland: Central Bank Swaps to the Eurozone, Poland, and Hungary, 2008–2009	Switzerland–Multiple 2008–09	French 2023a
United States: Swaps to the Bank for International Settlements and Deutsche Bundesbank, 1967	US–BIS and Bundesbank 1967	Arnold 2023e
United States: FIMA Repo Facility, 2020	US FIMA Repo 2020	Kelly 2023
United States: Central Bank Swaps to Mexico, 1982	US–Mexico 1982	Swaminathan 2023b
United States: Swaps to Mexico, 1994	US–Mexico 1994	Swaminathan and Wiggins 2023
United States: Central Bank Swaps to the Eurozone, United Kingdom, and Canada, 2001	US–Multiple 2001	French 2023d
United States: Central Bank Swaps to 14 Countries, 2007–2009	US–Multiple 2007–09	French 2023b
United States: Central Bank Swaps to Five Countries, 2010–2011	US–Multiple 2010–11	French 2023c
United States: Central Bank Swaps to 14 Countries, 2020	US–Multiple 2020	Hoffner 2023d

Source: Authors' compilation.

Major Themes Gleaned from the Cases

Swaps and swap-like facilities have proved their effectiveness as an important part of central banks' LOLR function.

The five dominant reserve currency central banks (the Fed, ECB, Bank of Japan [BOJ], Bank of England [BoE], and PBOC) (see the Appendix for frequently cited central bank names and abbreviations) have used swap lines to improve overseas liquidity conditions (Horn et al. 2023). From the lending central bank's perspective, swap arrangements enable the Lender to provide liquidity in its currency outside its borders without taking on the unknown risks associated with foreign commercial banks. By lending to the borrowing central bank, the Lender assumes limited exposure, secured by the Borrower's currency, and relies on the borrowing central bank to manage the downstream lending to its domestic banks, which it usually regulates and supervises.

From the borrowing central bank's perspective, swap arrangements enable it to provide LOLR funding in a needed foreign currency without having to deplete its foreign reserves or buy on the open market (Fleming and Klagge 2010). Even if central banks do not already have standing arrangements in place, new swap agreements can be signed and deployed rapidly. If a Borrower cannot secure a swap line, repo lines offer an alternative for the needed currency with additional security for the Lender.

Since the GFC, central banks have introduced a growing number of standing swap and swap-like (repo) arrangements that provide broad liquidity safety nets to many countries in major reserve currencies.

These standing arrangements are important precautionary measures to support global financial stability infrastructure (Allen and Moessner 2010; Steil, Della Rocca, and Walker 2021). The leading example of this development is the uncapped standing multilateral network among the Federal Reserve, Bank of Canada (BoC), Bank of England, Bank of Japan, European Central Bank, and Swiss National Bank (SNB), which includes all Group of Seven (G-7) countries and dominant reserve currencies except the renminbi ("Major Bank Swap Network"). The agreements governing this network permit any participating central bank to draw uncapped amounts of any other participating bank's currency in exchange for its own. Participants can activate such standing arrangements quickly when credit strains develop to provide foreign currency liquidity downstream to banks. New temporary swap lines can also be implemented quickly since swap arrangements are embodied in relatively simple agreements.

Swaps require effective central bank cooperation for the benefit of the Lender and Borrower.

By their nature, swaps require cooperation between different central banks, typically to avoid a liquidity crisis in the Lender country's currency and in the Borrower country's financial institutions (Bahaj and Reis 2021). Lending central banks can also coordinate draws under their swap arrangements, with the downstream lending administered by the borrowing central bank.

This cooperation can benefit the Lender as well as the Borrower. For example, during the GFC, the Fed coordinated domestic dollar lending under the Term Auction Facility (TAF) with overseas dollar lending by the SNB and ECB (Sheets 2019). (See French 2023b and Runkel 2022.) Nathan Sheets, a former director of the Division of International Finance at the Fed, notes that the cooperation with the ECB gave the Fed more advanced information about the ECB's lending activities than if the ECB had just injected the liquidity using its reserves, thereby enhancing the Fed's monetary control (Sheets 2019).

Lending central banks can customize swap arrangements to manage the risk to their balance sheets.

Swap agreements among advanced economy countries tend to have fairly similar terms. However, we observe far more customization in swap arrangements that involve emerging market countries. The need for customization is heightened when the Borrower's currency is not fully convertible—that is, not freely usable for settlements of international transactions in the way the dominant reserve currencies are (as with the Argentinian peso or Brazilian real, for instance). In such cases, lending central banks introduced additional requirements to minimize their credit risk. In two swap arrangements, the Lender required the borrowing central bank to provide collateral in a currency other than the Borrower's currency. Sometimes, the Lender required additional collateral or top-up provisions. Other times, the Lender required some ex ante qualification before the Borrower could draw on the swap, such as requiring the Borrower to pursue a loan from the International Monetary Fund (IMF). Such terms limit the risk to the Lender. Without them, some of the swap arrangements in this survey may not have happened (see Key Design Decision No. 5, Administration, and Key Design Decision No. 9, Process for Utilizing the Swap Agreement).

Reserve currency central banks have increasingly used repurchase agreements to provide their currencies to a broader network of central banks while limiting their exposure to credit risk.

A lending central bank may limit the number of counterparties with whom it enters into swap arrangements for a number of reasons. Such limits may impact emerging market economies disproportionately since they are more likely to need reserve currencies in a liquidity crisis to collateralize swaps. Rather than offer a swap with customized features to manage their credit risk, lending central banks have increasingly used repos. With repos, the Lender can receive as collateral government securities denominated in its own currency instead of the counterparty's currency. Since the GFC, the ECB has established bilateral repo lines with a number of non–euro area European Union (EU) central banks. Recently, both the ECB and the Fed have adopted broad-based repo frameworks that can greatly expand other countries' access to euros and US dollars. These frameworks are potentially accessible to almost any country, providing quick access to reserve currencies. However, they are useful only to Borrower countries that have meaningful holdings of government debt issued by the reserve currency country.

Regional arrangements seek to fill the gaps in central bank swap lines but have not been fully tested.

Another development is the emergence of regional cooperative swap arrangements such as the Association of Southeast Asian Nations (ASEAN) Swap Arrangement (ASA), the Chiang Mai Initiative Multilateralization (CMIM), and the South Asian Association for Regional Cooperation (SAARC) frameworks. The ASA and CMIM are both multilateral swap facilities, which enable participating countries to collectively provide swaps to another participant during a crisis. The SAARC framework has accomplished a similar goal but through a single Lender, the Reserve Bank of India (RBI). To date, the SAARC framework has provided a relatively modest amount of support to various participants. The CMIM remains untested, and the ASA has yet to be renewed after expiring in 2021. See Key Design Decision No. 1, Purpose and Type (B), for further discussion.

Swaps are frequently accompanied by other related actions to address a crisis and can serve as a bridge to longer-term assistance from international organizations.

When a liquidity crisis begins in an advanced economy, the central bank in that country may use swaps to support the liquidity of its currency abroad, while taking other actions to promote liquidity at home, such as adjustments to discount window terms or the introduction of new lending facilities.

When the Borrower's country is an emerging market economy, swaps can prove vital by replenishing a foreign currency that has been depleted. A Lender's country may extend this lifeline because it has a close economic or financial relationship to the Borrower's country. Such loans can serve as a bridge as the Borrower's country seeks longer-term assistance from the IMF (see Key Design Decision No. 2, Part of a Package).

Key Design Decisions

The following sections discuss 16 Key Design Decisions. This series is different from other New Bagehot Project surveys in that every case involves at least two central banks, a Lender and a Borrower. To be consistent with other surveys, we take the perspective of the lending central bank, since the lending central bank is the liquidity provider. Taking that perspective, our main focus is on the Lender's motivations in establishing the swap arrangement and how it has handled the key design decisions. However, in several cases, the liquidity issues originated in the Borrower's economy. For these reasons, we also discuss the swap or repo arrangement from the borrowing central bank's perspective when relevant.

1. **Purpose and Type (A): Central banks enter into swaps and repos for several reasons, but since the Global Financial Crisis, they have increasingly used these facilities to address liquidity constraints in their currencies outside their borders.**

Central banks enter swap arrangements for five basic purposes: (i) to provide downstream liquidity in the Lender's currency to a foreign jurisdiction, effectively extending discount window or LOLR operations; (ii) to boost reserves of the Borrower; (iii) to respond to or encourage increased trade in the Lender's currency; (iv) to manage balance of payments; or (v) to serve as bridge funding for the Borrower to achieve a multilateral support package from international sources such as the IMF.

Since our survey focuses on crisis-fighting measures, all but two of the 23 cases we reviewed included liquidity provision as a goal; the two that did not still emphasize financial stability through supporting the borrowing central bank's reserves (Scandinavia–Iceland 2008 and China–Argentina 2014). Thirteen cases stated liquidity provision as their sole purpose. (See Figure 3.)

Central banks use swaps to provide downstream liquidity in the Lender's currency to banks in the Borrower's foreign jurisdiction, effectively extending the Lender's discount window or LOLR function.

Since the GFC, central banks have greatly expanded the use of swaps for liquidity provision. The overwhelming majority of the cases in our survey addressed liquidity strains at times of crisis or market stress or when such was imminent. Most of these swap lines sought to facilitate downstream lending in the Lender's currency to financial institutions in the borrowing central bank's jurisdiction (see Key Design Decision No. 10, Downstream Use of Borrowed Funds). The regional swap frameworks in three of our cases—ASA 1977, ASEAN +3 CMIM 2010, and SAARC 2012—were not entered into in close proximity to a crisis. However, ASA 1977 and ASEAN +3 CMIM 2010 anticipated the use of the resulting swap arrangements for liquidity purposes in a crisis.

Swaps enable the lending central bank to extend its role as LOLR internationally, with the borrowing central bank bearing the credit risk of the downstream lending to its domestic financial institutions. "What is novel is the presence of an intermediary: the recipient central bank doing the monitoring, picking the collateral, and enforcing repayment" (Bahaj and Reis 2022b, 1659). The Lender bears only the sovereign credit risk of the Borrower, a risk that participants typically consider to be negligible in the case of advanced economies. There are good reasons for the Lender to shift credit risk to the Borrower. The Borrower should have greater knowledge of the domestic financial institutions that it supervises and regulates. It should thus be in a better position to evaluate the solvency of the borrowing banks, the quality of their collateral, and the potential for moral hazard in ex ante bank risk-taking (Bahaj and Reis 2018).

During the GFC, the Fed established a new precedent by resurrecting swap lines, which had previously been used extensively during the Bretton Woods era for currency stability, as liquidity-providing crisis-fighting tools. Beginning in late 2007, the Fed extended swap lines to 14 central banks to "[provide] dollar funds that those central banks could lend in their jurisdictions" (Fed 2008, 33) and "improve liquidity in dollar term funding markets" (FOMC 2007a, 13-14). Many of those swap lines were drawn upon shortly after adoption and then repeatedly; some were never drawn upon but in aggregate were one of the Fed's most significant tools in combatting the crisis with a peak outstanding of USD 583 billion (Fed n.d.a). The Fed also reinstated five swap lines during the Sovereign Debt Crisis (SDC) (US–

Multiple 2010–11) and enhanced the terms of these lines during the COVID-19 pandemic, while reintroducing the other nine on a temporary basis (which had a peak outstanding of USD 449 billion between the 14 lines) (Perks et al. 2021). Other central banks also quickly recognized the utility of swaps. Other GFC-era arrangements included the Swiss National Bank's swaps with the ECB, National Bank of Poland (NBP), and the central bank of Hungary (Magyar Nemzeti Bank, or MNB). The ECB's swap with Danmarks Nationalbank (DNB) and the ECB's repos with the NBP and MNB are also in this category.

Some central banks have adopted swap lines as precautionary measures early on rather than in the midst of crises (Allen and Moessner 2010; Steil, Della Rocca, and Walker 2021). For example, the Eurozone and Sweden set up a swap in December 2007 "to provide liquidity in EUR in case of adverse developments" (ECB 2007, 1). Sweden first drew on the swap in 2009 (ECB 2009b). Other examples are the ECB's swap lines with the Central Bank of Ireland (CBIR) and the BoE in 2010. The ECB said of these lines that they would "[allow] pounds sterling to be made available to the Central Bank of Ireland as a precautionary measure, for the purpose of meeting any temporary liquidity needs of the banking system in that currency" (ECB 2010). The line was only minimally used (Wiggins 2023).

Central banks use swaps to boost reserves.

In six of our cases, central banks established swap lines as a precautionary measure to bolster the foreign reserves of a recipient central bank (China–Argentina 2014, Scandinavia–Iceland 2008, US–Mexico 1982, and US–Mexico 1994, as well as SAARC 2012 and US FIMA Repo 2020). This lending can help support a borrowing central bank as it pursues additional long-term assistance. In May 2008, the central banks of Denmark, Norway, and Sweden established bilateral euro swap lines with the Central Bank of Iceland (CBI). At the time, the CBI had inadequate foreign reserves relative to the liabilities of its country's banking system. It entered into the swap arrangements "to strengthen confidence in the Icelandic economy in the short term" and to stabilize the economy while the government addressed fundamental vulnerabilities (SIC 2010, 175). The lending central banks did not expect the CBI to draw upon the facilities, although it later did (SIC 2010, 1:175).

Central banks use swaps to manage their balance of payments issues.

Five of our cases emphasized the use of swaps to address balance of payments issues, typically for emerging market economies. The ASA's purpose was "to provide short-term foreign exchange liquidity support for member countries that experience balance of payment difficulties" (ASEAN 2005). The CMIM resolved "to address balance of payment and/or short-term liquidity difficulties in the ASEAN+3 region" (AMRO n.d.). These facilities were not used during the GFC or the COVID-19 pandemic (Hoffner 2023a; Hoffner 2023b; Steil, Della Rocca, and Walker 2021). The RBI launched the SAARC swap framework in 2012 to provide backstop financing to other South Asian countries facing "balance of payments and liquidity crises" (RBI 2012).

Central banks use swaps to promote trade in their currencies.

In four of our cases, trade-related goals were prominent. In 2011, China established a swap line with Mongolia to "facilitate bilateral investment and trade and safeguard regional financial stability" (PBOC 2012). This was one of five swap lines China established that year. By 2020, the PBOC had extended 31 swap lines (Perks et al. 2021). The PBOC uses swap lines to promote the renminbi as a trade currency (China Daily 2014). Scholars have concluded that these lines effectively promote Chinese trade (Liao and McDowell 2015; Perks et al. 2021). However, recent scholarship by Horn et al. (2023) demonstrates that countries (often with weak credit and low reserves) have used their swap lines with China to resolve persistent balance of payments issues rather than to address a shortage of renminbi liquidity.

Central banks may use swaps as a bridge to longer-term financing.

Swap lines also often served as a bridge to longer-term support from international sources such as the IMF or Bank for International Settlement (BIS). For example, Iceland in 2008 and Mexico in 1982 and 1994 used swap arrangements as bridge funding to stabilize their currencies until they could arrange longer-term financial assistance from the IMF or other sources. In addition, among our cases, seven other countries (Argentina, Hungary, Ireland, Poland, Sri Lanka, the Maldives, Mongolia) also received IMF assistance in close proximity to the swap arrangement or during its tenure.

Borrowing central banks may use swap lines for other than their intended purposes.

Whether drawn funds will be used as expected is a factor that lending central banks consider in deciding to enter into a swap arrangement: swap lines may be used for purposes other than those for which they were originally intended. Typically, swap agreements do not include limitations on the use of funds received from a swap draw; the Borrower determines how it will use the funds drawn (see Key Design Decision No. 10, Downstream Use of Borrowed Funds). During the GFC, the SNB announced that it would draw USD 60 billion under its swap line with the Fed to help finance purchases of assets from a major Swiss bank, an atypical use (FOMC 2008). When the Fed approved swaps with four emerging market central banks (Brazil, Korea, Mexico, and Singapore), the Fed included additional governance safeguards and tranching of draws to maintain a tighter rein on usage. The four central banks also had to disclose up front how they were going to use the swap funds before draws were approved (FOMC 2008). Thus, regardless of when a swap arrangement is entered into, or for what purpose, there is some risk that a borrowing central bank may use the swap to provide foreign currency liquidity to domestic banks or for other crisis-related assistance should a need arise.

Figure 3: Major Purposes of Swap and Repo Arrangements

Case	Provide Downstream Liquidity	Boost Reserves	Promote Cross-border Trade	Manage Balance of Payments
ASA 1977	Yes	No	No	Yes
ASEAN +3 CMIM 2010	Yes	No	No	Yes
China–Argentina 2014	No	Yes	Yes	No
China–Hong Kong 2009	Yes	No	Yes	No
China–Mongolia 2011	Yes	No	Yes	Yes
Eurozone–Denmark 2008	Yes	No	No	No
Eurozone–EUREP 2020	Yes	No	No	No
Eurozone–Hungary Repo 2008	Yes	No	No	No
Eurozone–Ireland and UK 2010	Yes	No	No	No
Eurozone–Poland Repo 2008	Yes	No	No	No
Eurozone–Sweden 2007	Yes	No	No	No
Eurozone–UK 2019	Yes	No	No	No
SAARC 2012	Yes	Yes	Yes	Yes
Scandinavia–Iceland 2008	No	Yes	No	No
Switzerland–Multiple 2008–09	Yes	No	No	No
US–BIS and Bundesbank 1967	Yes	No	No	No
US FIMA Repo 2020	Yes	Yes	No	No
US–Mexico 1982	Yes	Yes	No	Yes
US–Mexico 1994	Yes	Yes	No	No
US–Multiple 2001	Yes	No	No	No
US–Multiple 2007–09	Yes	No	No	No
US–Multiple 2010–11	Yes	No	No	No
US–Multiple 2020	Yes	No	No	No

Source: Authors' compilation.

Purpose and Type (B): We classify swap arrangements into three broad categories: bilateral swap arrangements, multilateral swap arrangements, and swap-like repo arrangements. Such arrangements can also be temporary or standing (not having an end date).

Bilateral Swap Arrangements

Bilateral swap arrangements involve two parties. They can be either *unidirectional*, allowing one party to be the Lender and the other the Borrower, or *reciprocal*, allowing either party to be the Lender or Borrower.[7] Reciprocal agreements are most likely to occur between advanced economies, such as the Major Bank Swap Network among the Fed and five other major central banks, which the parties made standing in 2013, or the SNB–ECB arrangement (ECB 2022c; Fed 2020f). Agreements between advanced economies and emerging market economies (or countries in similarly vulnerable economic positions) are more likely to be unidirectional. Examples include the SNB's lines with Poland and Hungary, and the Scandinavia–Iceland swaps (ECB 2022c; French 2023a; Hoffner 2023c). Bilateral and multilateral agreements may combine into de facto series or networks (see also Figure 4).

Multilateral Swap Arrangements (Series and Networks)

Multilateral swap arrangements involve three or more parties. When these arrangements have a particular focus on one party, we refer to them as a *swap series.* A series can be a *Borrower-focused series* (BF series) with multiple Lenders and a single Borrower, such as the Scandinavia–Iceland case. There are also *Lender-focused series* (LF series) with multiple Borrowers and a single Lender, as in the US–Multiple 2001 case. The SAARC swap framework is a Lender-focused arrangement among India and other members of the SAARC, in which India is the only Lender. It anticipates lending Indian rupees (INR), US dollars, or euros in exchange for the Borrower's currency. The related swap lines have been unidirectional and temporary, resulting in a Lender-focused series (Gupta 2023e).

Some swap arrangements involving three or more parties differ from series, and we call these *swap networks*, because each party has a relationship with all other parties and is potentially a Lender and a Borrower; this is the case with the Major Bank Swap Network. The two Asian multilateral swap arrangements, the ASA (among 10 ASEAN countries) and the CMIM (among 10 ASEAN countries, China [including Hong Kong], Japan, and Korea), coordinate lending of US dollars, yen, or euros by participating members in exchange for the Borrower's domestic currency (Hoffner 2023a; Hoffner 2023b).

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[7] Our language isn't always consistent with the language central banks use because they are not always consistent with each other. The ECB uses the term "reciprocal," as we do, to describe bilateral arrangements in which either party can be the Lender or Borrower (ECB 2022c). The Fed also describes its North American Framework Agreement (NAFA) bilateral currency swap lines with Canada and Mexico as a reciprocal currency arrangement. However, the Fed typically calls swap arrangements in which it is the Lender "dollar-liquidity swap lines," and arrangements in which it is the Borrower "foreign currency liquidity swap lines" without reference to the arrangement being unilateral or reciprocal (Fed 2020f). We have chosen the ECB's terminology for clarity and uniformity, and because its usage appears more common among other central banks and scholars.

Swap-like Repo Arrangements

Central banks have used swap-like repo arrangements, both bilateral agreements and broad-based repo frameworks, to expand swap eligibility beyond major advanced economies; these two major developments are likely to see more usage in the future. The ECB and Fed use repos to provide their currencies in exchange for their own government securities as collateral, instead of accepting the Borrower's currency. Their repo frameworks are available to broad groups of potential participants, representing a novel solution to the "gap" created by Lender nations' tendency to limit with which countries they enter into swaps. (See Key Design Decision No. 7, Eligible Institutions, for related discussion.) Repo arrangements, secured by high-quality collateral, provide more security for the lending central bank. Borrowers tend to prefer swap agreements because they allow access to a currency issued by another central bank using their own currency. They also tend to be cheaper for the Borrower (ECB 2020d).

Swaps and repos may be temporary or standing. Since the GFC, a growing number of standing facilities from the Fed and ECB provide coverage to a significant portion of the integrated financial system.

Standing arrangements have no specified end dates; however, in all cases that we observed, these arrangements were still reviewed periodically by the parties. Standing swap arrangements remain in effect and dormant even when not in use, sometimes for years; they allow for rapid deployment if needed. Standing agreements are more prevalent between central banks of advanced economies; they may be unidirectional or reciprocal.

Central banks usually announce that they have "activated" a standing agreement when they intend to begin using it again. Activation can mean several things but often indicates that the parties have taken some action, such as a draw under the line or the announcement of a corresponding downstream lending program, that moves the swap line closer to use.

The BoE and ECB activated their standing swap arrangement in 2019, during the United Kingdom's exit from the European Union, to enable the BoE to lend euros to UK banks on a weekly basis. The operation complemented an existing BoE weekly US dollar repo facility and a move to weekly sterling operations (BoE 2019; ECB 2019). The ECB said, "the activation marks a prudent and precautionary step by the Bank of England to provide additional flexibility in its provision of liquidity insurance" (ECB 2019).

In 2020, the Fed introduced the FIMA Repo Facility as a temporary facility. In July 2021, after two extensions, the Fed converted it into a standing facility and permitted any eligible country to borrow up to USD 60 billion (Fed 2021b). By specifying and publishing some basic terms of the agreements in a framework facility, the Fed may minimize its administrative time in determining eligibility, which is still considered on an individual basis. Such a framework also provides the Fed with flexibility in seeking to balance the availability of its currency in foreign countries with other factors such as the security of the counterparty's currency.

Although the FIMA Repo Facility has not yet been heavily used, government officials, market participants, and scholars have commented positively on the establishment of the framework (Goldberg and Ravazzolo 2021b). Scholarship reveals that precautionary measures like the FIMA Repo Facility and the ECB's EUREP helped reduce the premium paid by foreign agents in foreign exchange markets (Albrizio et al. 2022; Choi et al. 2022).

Even though the central banks of advanced economies have established more standing swap facilities, they continue to use temporary swap lines, even with other advanced countries. For example, the Fed established temporary unidirectional swap lines with nine advanced and emerging market central banks during the GFC and again during the COVID-19 crisis, but in each case allowed those lines to end after the crisis had passed (French 2023b; Hoffner 2023d). Agreements between advanced economies and emerging market economies are still mostly likely to be temporary and unidirectional, such as the SNB's lines with the NBP and the MNB (see ECB 2022c; French 2023b; Gupta 2023b; Gupta 2023a; Hoffner 2023d).

The PBOC's large swap network is mostly characterized by temporary swaps of two to three years' tenure, which are often renewed periodically. However, in 2022, the PBOC expanded and converted its long-standing agreement with the Hong Kong Monetary Authority (HKMA) into its first standing swap agreement saying, "It will provide more stable and longer-term liquidity support for Hong Kong market, better supporting the development of Hong Kong as an international financial center and fostering the development of the offshore [renminbi] RMB market in Hong Kong" (PBOC 2022, 23).

Figure 4: Swap Types and Features

Types and Features	Examples of Facilities
Bilateral Swap Arrangements (two parties)	
Unidirectional: One-way swap. An arrangement pursuant to which central bank A lends its currency to central bank B in exchange for an equivalent amount of B's currency that it deposits with A to be held as collateral. At the end of the swap, B returns to A the borrowed currency with interest, and A returns to B the collateral currency without any interest.	Eurozone–Sweden 2007; Eurozone–Denmark 2008; Eurozone–Ireland and UK 2010
Reciprocal: Two-way swap. An arrangement that allows either central bank to borrow the other central bank's currency. Again, only the Borrower pays interest at the end of the swap.	US–BIS and Bundesbank 1967; China–Hong Kong 2009; Eurozone–UK 2019
Multilateral Swap Arrangements (three+ parties)	
Multilateral Series	
Swap arrangements may involve more than two parties to provide greater support during a crisis; this often occurs with countries that are geographically, economically, or culturally close. We define a series as a package of bilateral swap lines characterized as either Borrower focused or Lender focused. The underlying bilateral swaps can be reciprocal or unidirectional.	
Borrower-Focused Series. An arrangement in which several central banks coordinate bilateral swap lines to one borrowing central bank. For instance, central banks A and B agree to coordinate lending their currencies to central bank C in exchange for its currency.	Scandinavia–Iceland 2008
Lender-Focused Series. An arrangement in which central bank A agrees to lend its currency to central banks B and C in exchange for their currencies.	Switzerland–Multiple 2008–09; SAARC 2012
Multilateral Networks	
The key element of this type of swap arrangement is mutualization. It is an arrangement among three or more parties in which any party can borrow currency from any of the other parties by exchanging its own currency, which is held as collateral. Whereas a series consists of bilateral swap lines from either one Lender to multiple Borrowers or multiple Lenders to one Borrower, a network maintains intersecting swap lines between all (or most) parties. We define two types of networks.	
Multilateral Network. A network where central banks A, B, and C all have swap line agreements with each other, and any party can borrower currency from any other, forming a **web** of bilateral swap lines. The Major Bank Swap Network among the Fed and five other major central banks is an example of a multilateral network.	US–Multiple 2020
Commitment Pool Network. A network where there is a multilateral agreement in which all parties commit to provide currency into a common pool (not a fund). When participating party A requests a foreign currency, the other parties, B, C, and D, collectively extend swaps to the borrowing central bank in agreed-upon funding proportions.	ASA 1977; ASEAN +3 CMIM 2010
Swap-like Repo Facilities	
Bilateral Foreign Currency Repos. A repo arrangement under which central bank A lends its currency to central bank B in exchange for an equivalent amount of acceptable high-quality securities usually denominated in A's currency. At the end of the repo, B returns to A its borrowed currency with an agreed upon amount of interest, and A returns to B its collateral without any interest. Repo arrangements may be ad hoc unilateral or reciprocal arrangements or may be pursuant to a foreign currency repo framework; the ECB has employed both.	Eurozone–Hungary 2008; Eurozone–Poland 2008
Foreign Currency Repo Framework. A framework that facilitates the sponsoring central bank A's entering into bilateral repo agreements with other central banks, B and C, that wish to borrow A's currency in exchange for high-quality securities, usually denominated in A's currency. The bilateral agreements are independently negotiated within certain standard terms of the framework, such as maximum borrowing amount or acceptable collateral. The frameworks are intended to have a broad reach.	US FIMA Repo 2020; Eurozone–EUREP 2021

Source: Authors' analysis.

Major lending central banks—the Fed, ECB, PBOC, and SNB—take a multitiered approach to the use of swaps and repos, with different terms for different types of Borrowers.

Major lending central banks employ the types of swap and swap-like facilities in various ways with different counterparties to provide liquidity in their currencies outside their borders (and to secure foreign currency). The different types of facilities enable major lending central banks to be more nuanced in how they engage with different counterparties to provide liquidity and protect their balance sheets, and they use these facilities in different ways. For Borrower countries that need a liquidity facility, reviewing the way a Lender country has deployed its liquidity tools can help it assess the likelihood that it might be able to secure a particular type of facility. Figure 5 shows how the lending central banks examined in our cases have deployed the different facilities.

Figure 5: Swap or Repo Line: Type and Currency

Case	Swap/Repo Arrangement	Type	Currency Lent	Eligible Collateral
ASA 1977	ASA 1977	Commitment pool network	USD, EUR, JPY	ASEAN currencies
ASEAN +3 CMIM 2010	ASEAN +3 CMIM 2010	Commitment pool network	USD, ASEAN+3 currencies	ASEAN +3 currencies
China–Argentina 2014	China–Argentina 2014	Bilateral	RMB, USD	ARS
China–Hong Kong 2009	China–Hong Kong 2009	Bilateral	RMB	HKD
China–Mongolia 2011	China–Mongolia 2011	Bilateral–reciprocal	RMB	MNT
Eurozone–Denmark 2008	Eurozone–Denmark 2008	Bilateral–unidirectional (LF series)	EUR	DKK
Eurozone–EUREP 2020	Eurozone–EUREP 2020	Repo framework	EUR	EUR securities[A]
Eurozone–Hungary Repo 2008	Eurozone–Hungary 2008	Bilateral–unidirectional (LF series)	EUR	EUR securities
Eurozone–Ireland and UK 2010	Eurozone–Ireland 2010	Bilateral–unidirectional	GBP	EUR
	Eurozone–UK 2010	Bilateral–unidirectional	GBP	EUR
Eurozone–Poland Repo 2008	Eurozone–Poland 2008	Bilateral–unidirectional (LF series)	EUR	EUR securities
Eurozone–Sweden 2007	Eurozone–Sweden 2007	Bilateral–unidirectional (BF series)	EUR	SEK
Eurozone–UK 2019	Eurozone–UK 2013	Bilateral–reciprocal (LF series)	EUR	GBP
SAARC 2012	India–Bhutan 2013	Bilateral–unidirectional (LF series)	USD, EUR, INR	BTN
	India–Sri Lanka 2015	Bilateral–unidirectional (LF series)	USD, EUR, INR	LKR
	India–Maldives 2019	Bilateral–unidirectional (LF series)	USD, EUR, INR	MVR
Scandinavia–Iceland 2008	Denmark–Iceland 2008	Bilateral–unidirectional (BF series)	EUR	ISK
	Norway–Iceland 2008	Bilateral–unidirectional (BF series)	EUR	ISK
	Sweden–Iceland 2008	Bilateral–unidirectional (BF series)	EUR	ISK
Switzerland–Multiple 2008–09	Switzerland–Eurozone 2008	Bilateral–unidirectional (LF series)	CHF	EUR
	Switzerland–Poland 2008	Bilateral–unidirectional (LF series)	CHF	EUR
	Switzerland–Hungary 2009	Bilateral–unidirectional (LF series)	CHF	EUR
US–BIS and Bundesbank 1967	US–Bundesbank 1962	Bilateral–reciprocal (LF series)	USD, DEM	DEM, USD
	US–BIS 1965	Bilateral–reciprocal (LF series)	USD, DEM	DEM, USD[B]
US FIMA Repo 2020	US FIMA Repo 2020	Repo framework	USD	US Treasuries
US–Mexico 1982	US–Mexico 1967	Bilateral–reciprocal	USD	MXN
	US–Mexico 1982 (Fed/UST)[C]	Bilateral–unidirectional	USD	MXN
	US–Mexico 1982 (UST)[C]	Bilateral–unidirectional	USD	MXN
US–Mexico 1994	US–Mexico 1994 (NAFA)[D]	Multilateral network	USD	MXN
	US–Mexico 1994 (TSL)	Bilateral–unidirectional	USD	MXN
	US–Mexico 1995 (AP)	Bilateral–unidirectional	USD	MXN
US–Multiple 2001	US–Canada 2001	Bilateral–reciprocal (LF series)	USD	CAD
	US–Eurozone 2001	Bilateral–reciprocal (LF series)	USD	EUR
	US–UK 2001	Bilateral–reciprocal (LF series)	USD	GBP

Case	Swap Line	Type	Currency Lent	Eligible Collateral
US–Multiple 2007–09	US–Eurozone 2007	Bilateral–unidirectional (LF series)[E]	USD	EUR
	US–Switzerland 2007	Bilateral–unidirectional (LF series)[E]	USD	CHF
	US–Australia 2008	Bilateral–unidirectional (LF series)	USD	AUD
	US–Brazil 2008	Bilateral–unidirectional (LF series)	USD	BZR
	US–Canada 2008	Bilateral–unidirectional (LF series)	USD	CAD
	US–Denmark 2008	Bilateral–unidirectional (LF series)	USD	DKK
	US–Japan 2008	Bilateral–unidirectional (LF series)[E]	USD	JPY
	US–Korea 2008	Bilateral–unidirectional (LF series)	USD	KRW
	US–Mexico 2008	Bilateral–unidirectional (LF series)	USD	MXN
	US–New Zealand 2008	Bilateral–unidirectional (LF series)	USD	NZD
	US–Norway 2008	Bilateral–unidirectional (LF series)	USD	NOK
	US–Singapore 2008	Bilateral–unidirectional (LF series)	USD	SGD
	US–Sweden 2008	Bilateral–unidirectional (LF series)	USD	SEK
	US–UK 2008	Bilateral–unidirectional (LF series)[E]	USD	GBP
US–Multiple 2010–11	US–Canada 2010	Bilateral–unidirectional (LF series)[F]	USD	CAD
	US–Eurozone 2010	Bilateral–unidirectional (LF series)[F]	USD	EUR
	US–Japan 2010	Bilateral–unidirectional (LF series)[F]	USD	JPY
	US–Switzerland 2010	Bilateral–unidirectional (LF series)[F]	USD	CHF
	US–UK 2010	Bilateral–unidirectional (LF series)[F]	USD	GBP
US–Multiple 2020	US–Canada 2013	Multilateral network	USD	CAD
	US–Eurozone 2013	Multilateral network	USD	EUR
	US–Japan 2013	Multilateral network	USD	JPY
	US–Switzerland 2013	Multilateral network	USD	CHF
	US–UK 2013	Multilateral network	USD	GBP
	US–Australia 2020	Bilateral–unidirectional (LF series)	USD	AUD
	US–Brazil 2020	Bilateral–unidirectional (LF series)	USD	BZR
	US–Denmark 2020	Bilateral–unidirectional (LF series)	USD	DKK
	US–Korea 2020	Bilateral–unidirectional (LF series)	USD	KRW
	US–Mexico 2020	Bilateral–unidirectional (LF series)	USD	MXN
	US–New Zealand 2020	Bilateral–unidirectional (LF series)	USD	NZD
	US–Norway 2020	Bilateral–unidirectional (LF series)	USD	NOK
	US–Singapore 2020	Bilateral–unidirectional (LF series)	USD	SGD
	US–Sweden 2020	Bilateral–unidirectional (LF series)	USD	SEK

Notes: LF: Lender-focused; BF: Borrower-focused; Fed/UST: Federal Reserve/US Treasury; UST: US Treasury: NAFA: North American Framework Agreement; TSL: Temporary Swap Line; AP: assistance package of 1995.

[A] EUREP will accept collateral consisting of euro-denominated marketable debt securities issued by euro area central governments and supranational institutions.

[B] Although only Deutsche marks were used during the period of our case, the swap line authorized several currencies as collateral: Deutsche marks, Austrian schillings, Belgian francs, French francs, Italian lire, Netherlands guilders, Swedish kronor, Swiss francs, Canadian dollars, Japanese yen, and pounds sterling.

[C] UST represents swaps established by the US Treasury, funded through the Exchange Stabilization Fund, and administered by the Federal Reserve Bank of New York.

[D] The US–Mexico NAFA is part of a multilateral network with the BoC.

[E] These BoE, BOJ, ECB, and SNB swap lines with the Fed became bilateral–reciprocal in April 2009.

[F] These BoC, BoE, BOJ, ECB, and SNB swap lines became a multilateral network on November 30, 2011.

Source: Authors' analysis.

The Federal Reserve Approach

The Fed has adopted a three-tiered approach to swaps, as shown in Figure 6; it uses standing swaps, temporary swaps, and repos. The Fed considers the swap lines to "have helped to ease strains in financial markets and mitigate their effects on economic conditions . . . [they] support financial stability and serve as a prudent liquidity backstop" (Fed 2020g).

Since 2013, the Fed has used standing reciprocal swaps, which can be activated at any time, with five other major advanced-economy central banks, through the Major Bank Swap Network. During global crises, it also established temporary US dollar–providing swaps with nine other central banks, five that were from small advanced-economy countries and four that were from emerging market–economy countries (Brazil, Korea, Mexico, and Singapore). And since 2020, it has made a repo facility available to virtually any central bank that has the required US dollar–denominated collateral. Also see Key Design Decision No 7, Eligible Institutions, for further discussion of the Fed's approach.

Figure 6: Federal Reserve Liquidity-Providing Facilities as of March 31, 2020

Facility	Type	Counterparties
Standing swap arrangements[A]	Reciprocal	BoC, BOJ, BoE, ECB, SNB
Temporary swap arrangements	Unidirectional US dollar providing	Reserve Bank of Australia, BCB, DNB, BOK, Banxico, RBNZ, Norges Bank, MAS, Riksbank
FIMA Repo Facility[B]	Broad-based repo framework supporting bilateral unidirectional US dollar providing repo agreements	Central banks from 30+ countries[C]

[A] Part of the Major Bank Swap Network, a multinational network among six major central banks providing for uncapped borrowings in any counterparty's currency in exchange for the first party's own currency.

[B] The participant can request up to USD 60 billion in exchange for posting an equivalent amount of US Treasury bonds as collateral.

[C] The Fed does not disclose the names of FIMA participants, but as of 2020, roughly 30 of 170 eligible foreign agencies had enrolled.

Sources: Authors' compilation; Kelly 2023 (c).

The ECB Approach

The ECB has four types of facilities: unidirectional swaps; reciprocal swaps; bilateral repos; and, since 2020, bilateral repos under its temporary EUREP facility (ECB 2022c). As shown in Figure 7, it currently has unidirectional agreements with three central banks (Denmark, Poland, and Sweden). It also has reciprocal swap arrangements with six central banks; five of these are part of the Major Bank Swap Network, and the other is an agreement with the PBOC. All of the swap lines except for those with the National Bank of Poland and PBOC are standing arrangements (ECB 2022c).

As with the Fed, a key consideration for the ECB is also its willingness to accept the counterparty's currency as collateral. As an alternative, the ECB has established bilateral repos with six countries (Albania, Andorra, Hungary, North Macedonia, Romania, and San Marino); all are unidirectional and temporary. It also has established bilateral repos under its EUREP facility but does not disclose those participants (ECB 2022c).

Figure 7: ECB Liquidity-Providing Facilities as of December 2022

Facility	Type	Counterparties
Unidirectional swap arrangements	Standing	DNB, Riksbank
	Temporary	NBP
Reciprocal swap arrangements [A]	Standing	BoC, BOJ, BoE, ECB, SNB
	Temporary	PBOC
Bilateral unidirectional repo arrangements	Temporary	Bank of Albania, Andorran Financial Authority, MNB, National Bank of Northern Macedonia, National Bank of Romania, Central Bank of the Republic of San Marino
EUREP framework supporting bilateral unidirectional repo agreements[B]	Temporary	Unknown[C]

[A]-Part of the Major Bank Swap Network, a multinational network among six major central banks providing for unlimited borrowings in any counterparty's currency in exchange for the first party's currency.

[B]-The ECB negotiates the bilateral agreement individually with the counterparty; euros can be borrowed in exchange for posting an equivalent amount of high-quality euro-denominated government securities.

[C]- The ECB does not disclose the names of EUREP participants, but several countries have announced their participation.

Sources: Author compilation; ECB 2022c.

Membership in the European Union does not appear to dictate which form of liquidity facility the ECB chooses to deploy with a counterparty, and the ECB has on occasion switched from a repo to a swap arrangement with the same counterparty. For example, in 2008, the central banks of Hungary and Poland (both countries are members of the EU but not the euro area) each requested a swap agreement from the ECB. After much debate, the ECB instead offered temporary bilateral repo agreements, largely out of concern for the ECB's balance sheet. As the then–director for market operations explained, the "basic reason for doing repos instead of swaps was credit risk" (Spielberger 2023, 882). Repos were seen as a compromise between offering some support and protecting the bank (Spielberger 2023). In January 2010, the ECB and MNB converted half of the value of the MNB's EUR 5 billion repo facility into a currency swap facility. However, in July 2020, in response to the COVID-19 pandemic, the parties entered into a new EUR 4 billion repo facility (Gupta 2023b). In contrast, in March 2022, the ECB and NBP entered into a EUR 10 billion swap arrangement, the same amount as their previous bilateral repo (Gupta 2023b).

The PBOC Approach

Our cases include PBOC swaps with Argentina, Hong Kong, and Mongolia. Many of the terms of these swaps are undisclosed, but we know at least that the PBOC's swap line with Mongolia is structured as a reciprocal arrangement (Arnold 2023c). China has established approximately 40 swap lines since 2008 (Horn et al. 2023).

The PBOC approach differs from the Fed and ECB approaches in several ways. The PBOC's swap lines (i) cover a larger and more diverse group of counterparties; (ii) address trade and settlement issues, not liquidity issues; and (iii) support the PBOC's drive to internationalize the renminbi (Bahaj and Reis 2022a). Many of these swap lines go unused for years.

Nonetheless, Horn et al. (2023) finds that the PBOC's lines are "increasingly used as a financial rescue mechanism, with more than USD 170 billion in liquidity support extended to crisis countries [mostly distressed countries with low liquidity ratios], including repeated rollovers of swaps coming due" (1).

The SNB Approach

During the GFC, the SNB entered into swaps with the ECB and the central banks of Hungary and Poland, which had strong demand for Swiss francs from banks in their jurisdictions. The SNB and ECB amended an earlier unidirectional arrangement that enabled the SNB to borrow euros, making it reciprocal. The SNB then entered temporary unidirectional swap agreements to lend euros to the Hungarian and Polish central banks, mimicking existing Fed and ECB swaps with emerging market economies. The SNB's swaps with the Hungarian and Polish central banks required euros as collateral, rather than the two borrowing central banks' local currencies (French 2023a).

Customization of Swap Arrangements to Use a Third Currency

In some swap arrangements, the lending central bank provided (or could have provided) a currency not its own (see Scandinavia–Iceland 2008 and SAARC 2012). Also, in some cases, the borrowing central bank was required to provide a currency as collateral that was not its own (see, for example, Switzerland–Multiple 2008–09). This type of swap can be useful when the Borrower does not have direct access to the third-party currency central bank or if the Lender is reluctant to accept the Borrower's currency but prefers to continue with a swap rather than a repo arrangement. We see this occurring in agreements with emerging markets. For example, in 2015, India entered into a bilateral swap agreement with Sri Lanka pursuant to the SAARC swap framework. Sri Lanka was eligible to request US dollars, Indian rupees, or euros in exchange for its rupee (Gupta 2023e). Switzerland also provided euros to Hungary and Poland in exchange for euros instead of forints or złoty. The multilateral networks, the ASA and ASEAN, also contemplate that a participant can request US dollars, euros, or yen in exchange for its currency.

2. **Part of a Package: Lending central banks often take other measures to address liquidity stresses in their currencies, both domestically and internationally. Borrowing central banks often seek additional aid through swaps with other countries, other sources of foreign liquidity, or other support from lending central banks.**

Lending central banks commonly take additional measures to address liquidity stress in their currencies by offering swaps in multiple countries and by taking measures to shore up domestic liquidity.

While there were many Borrowers in our series, there were fewer Lenders (excluding the multilateral framework arrangements). Lenders comprised the central banks of China, Denmark, Eurozone, India, Norway, Sweden, Switzerland, the UK, the US, and West

Germany.[8] All except the PBOC had additional swaps or swap-like facilities in place to address concerns about the liquidity of their currencies in other countries at the same time as our case. These additional swaps can be characterized as (1) standing swap lines implemented as backup liquidity facilities, or (2) temporary swap lines implemented to address a specific crisis incident. This prevalence is consistent with the growing use of swap lines as liquidity vehicles observed by scholars and policymakers since the GFC (Allen and Moessner 2010; Bahaj and Reis 2018).

Among our cases, a crisis was present or imminent in 20 of 23 cases including cases occurring during the GFC or COVID-19 pandemic, when broad-based disturbances were impacting both parties to the swap to different degrees. In the majority of these cases, the impetus for the swap was the stress in the markets for the Lender's currency creating demand; the swaps served to provide their currencies outside their countries. In all of these cases, the lending central banks (the Fed, ECB, BoE, DNB, Norges Bank, Riksbank, and SNB) also increased domestic lending, sometimes in foreign currencies (such as when the ECB implemented domestic lending in pounds sterling and US dollars utilizing its swap arrangements) (see Figure 8). Lenders also took other measures to address distress in their financial systems and money markets, including broad-based emergency lending programs, credit guarantees, asset purchases, quantitative easing, capital injections, restructurings, and rule changes.[9]

The PBOC cases are different from the other cases in our survey because China generally establishes swaps not to address financial stability or liquidity concerns but to support a Borrower country's foreign reserves or to promote China's trade and investment settlement (Horn et al. 2023). Thus, it tends to set up swaps in combination with loans from government-owned banks and commodity prepayment facilities, which increasingly provide "a financial rescue mechanism" used by distressed countries with low liquidity ratios and low credit ratings to bolster gross reserves (Horn et al. 2023, 1). Nevertheless, as our China–Argentina 2014 case demonstrates, borrowing central banks may still be able to use these swap lines directly or indirectly to provide lending to their domestic firms (Arnold 2023c). Although China's support is small compared to that of the Fed and other major central banks from advanced countries, it has particular import for emerging and developing economies, which often find it difficult to secure swap lines with advanced economy nations (Horn et al. 2023).

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[8] In our US–BIS and Bundesbank 1967 case, it appears that the US drew under a reciprocal swap line with the Bundesbank and, in an unusual move, permitted the Bundesbank to loan downstream the dollars it provided as collateral to address constraints in the Eurodollar market. We have not discovered an explanation for why the Bundesbank was not the borrowing party, the only difference being that as the Borrower, the US paid interest when the transaction was unwound. See Arnold (2023e).

[9] For discussions of various programs implemented during crisis time to address wide-ranging disturbances, see Rhee et al. (2020) on market liquidity programs during the GFC and before; Rhee, Engbith et al. (2022) on market support programs during the COVID-19 crisis; Rhee, Oguri et al. (2022) on broad-based capital injection programs; McNamara et al. (2022) on blanket guarantee programs; and Wiggins et al. (2022) on broad-based emergency liquidity programs.

Borrowing central banks commonly seek additional aid through swaps with other countries, other support from lending central banks, or other sources of foreign liquidity to stabilize their financial markets.

There were 25 unique known borrowing central banks among our cases,[10] and eight cases involve situations where the stresses in the markets for the Lender's currency were largely contained to the Borrower nation. However, Borrower countries that were not advanced economies were less likely to have additional swap arrangements in place and were more likely to rely on other sources of foreign liquidity. For example, several countries that did not have swap arrangements with the Fed used other means to address US dollar shortages during the GFC. Russia was able to tap its substantial foreign currency reserves; India arranged a US dollar swap line with the Bank of Japan; and Hungary and Iceland received IMF standby arrangements (Allen and Moessner 2010)

In our cases, six Borrowers had no additional swap arrangements in place (Argentina, Brazil, Mongolia, Iceland, Bhutan, and the Maldives). Excluding the six central banks of the Major Bank Swap Network, eight borrowing central banks in our cases maintained swap lines with countries other than the Lender country during the period of the case, as shown on Figure 8.

The MNB and NBP each had swap lines with the SNB during the GFC (Switzerland–Multiple 2008–09) in addition to the repos with the ECB. The DNB and Riksbank had swap lines with the Fed in addition to those with the ECB; during the GFC, the DNB and Riksbank also established swap lines to provide euros to the central bank of Latvia, and the Riksbank had a separate swap line to provide Swedish krona to the central bank of Estonia (Allen and Moessner 2010; Gupta 2023c; Gupta 2023d). Sri Lanka and the Maldives had programs with the IMF, and Sri Lanka also had a swap line with the PBOC and was a participant under the FIMA Repo Facility (Gupta 2023e).

In 1982, Mexico maintained swap lines with France, Israel, and Spain, and in 1994, with Canada as part of the trilateral North American Framework Agreement (NAFA) among the US, Mexico, and Canada. In 1994, while the US chose to provide short-term swaps to bridge Mexico to a multilateral aid package funded by the IMF and US, it chose not to utilize the NAFA that was part of a trade deal among the three countries (Swaminathan and Wiggins 2023). The swift utility with which a swap line could be implemented permitted the US to maintain in place its carefully negotiated NAFA swap agreement while providing targeted liquidity under a new swap line. In 1967, the BIS also had a swap line with the SNB for Swiss francs (Arnold 2023e).

[10] Excluded from these totals are participating central banks in the CMIM, ASA, FIMA Repo Facility, EUREP, as well as the BIS. In theory, any participating central bank involved in the CMIM and ASA can act as a lender or borrower. The FIMA repo (US FIMA Repo 2020) or the EUREP (Eurozone–EUREP 2020) frameworks do not disclose their users. The BIS was also a borrower in the 1967 year-end operations with the Fed (US–BIS and Bundesbank).

Lending central banks are more likely to provide additional nonswap aid to Borrower countries when there is a strong regional economic, trade, or cultural connection between the parties.

With one exception, the Brexit case (Eurozone–UK 2019), the Lender country provided various nonswap aid to the Borrower country, which could be characterized as a regional hegemon aiding a peripheral economy—for example, US to Mexico (in 1982 and 1994), India to Bhutan and Sri Lanka, and China to Mongolia and Hong Kong. In these cases, the special relationship between the countries does seem to be one important impetus supporting the aid. For example, India, Bhutan, and Sri Lanka are members of the SAARC, which was founded for, among other things, mutual trust, mutual assistance, and strengthening cooperation and development of the peoples of South Asia, "which are bound by ties of history and culture" (SAARC 2020). Both Bhutan and Sri Lanka received swap lines from India under the SAARC swap framework (under which India has taken on a leadership role as the sole funder) but also received additional assistance from India in the form of special swaps, deferments, and credit lines (Gupta 2023e).

A second important factor present in some cases is the Lender's desire to avoid spillover effects, in other words, to contain the stress in the Lender's currency to the borrowing country and not have it transmitted to the lending country. For example, in discussions with the IMF and EU about the UK's contribution to a bailout for Ireland (in addition to its swap arrangement with the EBC for Ireland's benefit), then–UK Chancellor of the Exchequer George Osborne commented, "We are doing this because it is overwhelmingly in Britain's national interest that we have a stable Irish economy and banking system" (BBC News 2010). There were significant interconnections between the UK and Ireland banking systems (Wiggins 2023).

Figure 8: Part of a Package

Case	Central Banks the Lender Also Had Swap Lines With	Lender Expanded Domestic Liquidity Facilities	Lender Provided Nonswap Aid to Borrower	Central Banks the Borrower Also Had Swap Lines With	Borrower Received Third-Party Aid
ASA 1977	N/A[A]	N/A	N/A	N/A	N/A
ASEAN +3 CMIM 2010	N/A[A]	N/A	N/A	N/A	N/A
China–Argentina 2014	28[B]	No	Multiple measures[C]	None	IMF: USD 57 billion
China–Hong Kong 2009	6[B]	No	Multiple measures[D]	CMIM[D]	None
China–Mongolia 2011	14[B]	No	No information	None	IMF: USD 434 million
Eurozone–Denmark 2008	Fed, SNB, Riksbank	Yes	Yes	Fed, CBI, LB	None
Eurozone–EUREP 2020	15[E]	Yes	N/A	N/A	N/A
Eurozone–Hungary Repo 2008	Fed, SNB, Riksbank, DNB	Yes	None	Swap line with SNB	IMF: SDR 7.6 billion; EU: EUR 5.5 billion[F]
Eurozone–Ireland and UK 2010	Fed, SNB, BOJ, BoC[G]	No	UK loan of GBP 3.2 billion to Ireland[H]	None[G]	IMF, EU, Sweden, Denmark: EUR 85 billion[H]
Eurozone–Poland Repo 2008	Fed, SNB, DNB, Riksbank	Yes	None	Swap line with SNB	IMF: USD 20.6 billion
Eurozone–Sweden 2007	Fed, SNB	Yes	Yes	Fed, CBI, LB, EP	None
Eurozone–UK 2019	BOJ, BoC, SNB, ECB, Fed[I]	Yes	None	BOJ, BoC, SNB, ECB, Fed[I]	None
SAARC 2012	BOJ	Yes	Multiple measures[J]	Varies[K]	Multiple measures[L]
Scandinavia–Iceland 2008	Multiple[M]	Yes	Multiple loans[N]	None	IMF: USD 2.1billion
Switzerland–Multiple 2008–09	None	Yes	None	Multiple[O]	None
US–BIS and Bundesbank 1967	15[P]	N/A	None	BIS: swap with the SNB	N/A
US FIMA Repo 2020	14[Q]	Yes	N/A[Q]	Multiple[Q]	N/A
US–Mexico 1982	15[P]	No	Prepayments[R]	France, Israel, Spain	Multiple measures[S]
US–Mexico 1994	15[P]	No	Security guarantees	Canada[T]	IMF, BIS, Canada: USD 29 billion[T]
US–Multiple 2001	Canada and Mexico[U]	Yes	None	None	None
US–Multiple 2007–09	Canada, Mexico[U]	Yes	None	Multiple[Q]	None
US–Multiple 2010–11	Canada, Mexico[U]	No	None	Multiple[Q]	None
US–Multiple 2020	Canada, Mexico[U]	Yes	None[Q]	Multiple[Q]	None

Notes: N/A: not applicable; EP: Eesti Pank, the central bank of Estonia; LB: the Latvijas Banka, the central bank of Latvia. The second and fifth columns show any central banks with which the Lender and Borrower also had swap lines in close proximity to the time of signing the focus swap or repo arrangement.

[A] As parties to a multilateral network swap facility, the participating countries can be both Lenders and Borrowers; many of the participants also have bilateral swaps with other facility participants and/ or with other nonparticipant countries in the region.

[B] Since 2008, the PBOC has entered into swap arrangements with roughly 40 central banks, mostly in emerging market countries.

[C] China announced a Comprehensive Strategic Agreement between the two countries that encompassed 20 trade and investment deals, including the swap line.

[D] China announced 13 monetary and fiscal measures to assist Hong Kong, including the swap line. While the swap line was implemented during the GFC, it was not extended in response to a crisis-related renminbi shortage. As of 2010, the HKMA had repo lines in place with 10 central banks in the region and was also a participant in the CMIM.

[E] In 2020, the ECB maintained swap lines with nine central banks (BoC, BoE, BOJ, Bulgarian National Bank, Hrvatska Narodna Banka (Croatia), DNB, the Fed, the PBOC, and the SNB) and bilateral repo lines with the central banks of six countries (Albania, Hungary, Macedonia, Romania, San Marino, and Serbia). It does not disclose participants under the EUREP repo framework.

[F] Hungary drew SDR 7.6 billion from an IMF facility and EUR 5.5 billion from the EU's Balance of Payments facility.

[G] The lending central banks, the ECB and BoE, had swap arrangements with the Fed during this period. The Central Bank of Ireland, the ultimate intended borrower, had no swaps with any other countries during this period.

[H] The UK provided Ireland a GBP 3.2 billion loan as part of an EUR 85 billion bailout in which the EU, IMF, Sweden, and Denmark also participated.

[I] The ECB–BoE swap arrangement was part of the Major Bank Swap Network among the six major central banks (BoC, BoE, BOJ, ECB, Fed, and SNB) that was created in October 2013 as a standing multilateral network; each central bank could borrow any participant's currency in exchange for its own.

[J] In addition to SAARC swaps, Bhutan received from India over several years additional aid including a credit line aggregating INR 21 billion, an overdraft facility, and hydropower debt relief. Sri Lanka received a USD 1.4 billion support package from India that included a swap line, credit line for fuel imports, and deferment of dues at the Asian Clearing Union. We did not uncover any additional aid to the Maldives.

[K] Sri Lanka maintained swap lines with China and Bangladesh but also had access to the US FIMA Repo Facility.

[L] The IMF provided Sri Lanka USD 2.9 billion via the Extended Fund Facility and provided the Maldives with USD 28.9 million from its Rapid Credit Facility. Bhutan did not receive IMF aid other than general aid to all members.

[M] The central banks of Denmark, Norway, and Sweden entered into swap lines with the Fed in September 2008; the central banks of Sweden and Denmark also entered into swap lines with the ECB. Sweden extended swap lines to Estonia and Latvia, and Denmark extended a swap line to Latvia.

[N] After Iceland entered into a support agreement with the IMF (USD 2.1 billion) the governments of Denmark, Norway, Sweden, and Finland pledged loans totaling USD 2.5 billion to support the IMF relief.

[O] During the crisis period, the ECB had swap lines with Denmark, Switzerland, and the Fed and repo arrangements with Hungary and Poland.

[P] From 1962 on, the Fed maintained swap arrangements with up to 15 central banks for currency stabilization purposes. Most of these lines were permitted to expire by 1998.

[Q] During the GFC, the Fed had swaps with five other major central banks (BoC, BoE, BOJ, ECB, and SNB), which it reinstated in 2010 during the Sovereign Debt Crisis. After November 2011, each of the five borrowing central banks also had swaps with each other, and some also had swaps with other countries. In October 2013, these lines were converted to the Major Bank Swap Network. The Fed also had swaps with nine other central banks during the GFC and COVID-19 pandemic (Reserve Bank of Australia, BCB, DNB, BOK, Banxico, RBNZ, Norges Bank, MAS, and Riksbank); several of these counterparties also had swaps with other central banks. The Fed does not disclose participants under its FIMA repo framework; participants may have also had swap lines with or received other assistance from the US.

[R] The US Department of Energy prepaid USD 1 billion to Mexico for oil imports, and the US Department of Agriculture provided a credit of USD 1 billion through its Export-Import Bank and Commodity Credit Corporation.

[S] In August 1982, the BIS provided USD 925 million in short-term credits (funded by several industrial countries), a group of industrialized countries provided USD 2 billion in export credits, and a large consortium of commercial banks provided USD 5 billion in additional deposits. In December 1985, the IMF provided USD 3.75 billion in assistance under its Extended Fund Facility.

[T] The IMF provided USD 17.8 billion, the BIS provided USD 10 billion (funded by several industrial countries), and Canada provided USD 1.2 billion.

[U] Since 1994, the Fed has had a trilateral arrangement with the BoC and Banxico pursuant to the North American Framework Agreement.

Sources: Authors' compilation; Allen and Moessner 2010 (m); Boughton 2001 (s); ECB 2009a (o); ECB 2021a (e); Gupta 2023e (l); Hoffner 2023c (n).

Similarly, in 1994, Mexico suffered a currency crisis that threatened to impact Mexican subsidiaries of US banks. The US actively provided dollar liquidity through swaps to Mexico to sustain it until it could finalize a USD 48 billion IMF–US financial package to which the US contributed USD 20 billion. Then–US President Bill Clinton stated that the purpose of the package was to protect United States jobs, exports, and immigration interests, and address security concerns threatened by Mexico's liquidity crisis (Clinton et al. 1995; Swaminathan and Wiggins 2023).

In the Iceland case (Scandinavia–Iceland 2008), there is described a history of political and economic cooperation among the governments of the Nordic countries owing to the region's close geographic, cultural, and commercial ties (FSB 2016; Wajid et al. 2007).[11] Sweden, Norway, and Denmark agreed to provide euro swap lines to Iceland to support its vulnerable economy during the GFC. Following the collapse of the Icelandic banking system, Iceland drew on the swap line to secure trade in essentials before negotiating an IMF rescue package, to which the Scandinavian countries and Finland, another Nordic country, also contributed (Hoffner 2023c).

3. **Legal Authority: Swap arrangements are a core function of the central bank's mission. In all our cases, they were authorized under existing laws, usually by the highest governing body of the central bank, although in a few instances, approval from another body was required.**

All the central bank swaps arrangements we reviewed were authorized under the bank's existing legislative authority in its applicable jurisdiction; no new laws had to be adopted. Additionally, the authority to enter into swaps resides in the highest level of the central banks and is typically contained in the authorities for monetary policy or embedded in the authority to enter into foreign currency transactions.

For example, at the Fed, authority resides in the Board of Governors, which has decided to act through the Federal Open Market Committee (FOMC), a committee composed of governors of the central bank (FOMC 2020b; USG 1913). At the ECB, the Governing Council, the ECB's decision-making body, makes decisions regarding whether to enter into a foreign currency swap agreement (ECB 2021b; ECB 2022d).

In a few cases, approval or cooperation from another government entity in addition to the central bank, usually but not always the fiscal authority, was required (or if not required, a critical element to the arrangement). For example, the PBOC's enabling law gives it the authority to engage in "relevant financial activities," but most PBOC decisions (if not day-to-day monetary policy operations) must be approved by the State Council (Arnold 2023c). In another example, the DNB entered into its swap agreement with the ECB in 2008 only after negotiations with the Royal Bank Commissioner, as required by Section 2(3) of the Danish

[11] Recognizing the cross-border financial linkages throughout the region, the central banks of Denmark, Finland, Iceland, Norway, and Sweden signed a memorandum of understanding in 2003 establishing general principles of cooperation during financial crises (Suomen Pankki 2003). Owing to the severity of the Global Financial Crisis, the Nordic central banks found little use for the 2003 memorandum and its guidelines on information sharing and cooperation (FSB 2016).

Act on Foreign Exchange of 1988, which sets out foreign exchange policy for Denmark (Gupta 2023c). As described in Scandinavia–Iceland 2008, Norges Bank required approval from the Norwegian Ministry of Finance to establish a swap line with Iceland (Norges Bank 2009; Norwegian Government 2007). In the two cases of US assistance to Mexico, in 1982 and 1994, the Treasury independently extended swaps to the Bank of Mexico (Banxico) and also backstopped several Fed swaps through its Exchange Stabilization Fund[12] (Swaminathan 2023b; Swaminathan and Wiggins 2023).

The legal authority for countries to participate in the ASA, a multilateral swap arrangement, was based in the authorities of the central banks (and monetary authorities) of the participating countries. The original 1977 memorandum of understanding (MoU) on the ASEAN Swap Arrangements was signed by the central bank governors of the five originating countries (Indonesia, Malaysia, the Philippines, Singapore, and Thailand). The legal standing of the MoU may vary by country; in the case of the Philippines, the MoU was classified as a treaty and ratified through a senate resolution even though the MoU did not itself require ratification (Senate of the Philippines 1977).

The CMIM is not a standing fund or separate legal entity but is a contractual agreement among the participants that creates a multilateral swap arrangement in the form of a pooled reserve fund (AMRO n.d.; APT 2010; Han 2022). The agreement was signed by the finance ministers and central bank governors of the 13 participating ASEAN + 3 countries and the head of the Hong Kong Monetary Authority; thus, the arrangement relies on the authorities of the central bank and the fiscal authorities of the participating governments. (Ciorciari 2011). Under the agreement, each country commits to fund future requested draws under the agreement, up to predetermined amounts and subject to certain opt-out rights (Ciorciari 2011). The form of a MoU and a treaty were also considered but rejected in favor of the contract. However, the CMIM agreement does not necessarily hold the status of a law in each member country's domestic legal system, unlike a treaty would, and "was perceived to be accompanied with a lesser degree of commitment than would be the case with a treaty," which would have required ratification by each participating country (Han 2022, 736).

The CMIM's relatively light institutional design and legal basis reflects compromises made to comply with the members' differing legal systems. Initially, members agreed to structure the CMIM as a self-managed reserve pooling arrangement wherein participating central banks would transfer the legal ownership of a portion of their reserves to a newly created legal entity—a "virtual fund" (Han 2022). Although this model would have provided "greater financing certainty," it was ultimately abandoned owing to incompatibilities with certain members' legal systems (Han 2022). To expedite the establishment of the CMIM, members, therefore, opted for a contractual agreement with commitment letters. The CMIM has yet to be activated, and uncertainty of financing remains one of the agreement's biggest challenges.

[12] The US Treasury extends swap lines through the Exchange Stabilization Fund (ESF) (US Treasury n.d.). The swap facilities are administered by the Federal Reserve Bank of New York as fiscal agent for the US and the operating arm of the Fed. The president and Treasury secretary were also authorized to use the ESF to provide loans and credits to Mexico under section 10(a) of the Gold Reserve Act of 1934 (USG 1934) and under Title 31 under the United States Code, Section 5302 (Swaminathan 2023b; Swaminathan and Wiggins 2023; USG 1934, 5–6; US Treasury n.d.).

Han (2022), an official at AMRO (ASEAN + 3 Macroeconomic Research Office), the CMIM's surveillance unit, suggests that transforming the agreement into a treaty may reduce such uncertainty by enshrining the commitments into law.

The RBI initiated the SAARC framework (India is the sole funder) in 2012, pursuant to its authority under the 1934 Reserve Bank of India Act, but the government's Union Cabinet (the highest executive decision-making body) also had to approve the RBI's actions (GoI 1934; RBI 2022). All subsequent extensions and amendments to the SAARC framework also require Union Cabinet approval (RBI 2012).

4. Governance (A): Governance of swap and repo lines (and frameworks), including approval or amendment of arrangements and key terms (eligibility, amount, interest rate, collateral, duration of swaps), and approval of swap draws is done at the highest level of the central bank, although in some limited circumstances, approval by the government may also be required.

The authority to enter into swap and swap-like repo arrangements resides at the highest level of the central banks. At the Fed, this is the FOMC, a committee of the Board of Governors (FOMC 2020b; USG 1913). At the ECB, it is the Governing Council, the ECB's highest decision-making body. Typically, these bodies, and their equivalents at other central banks, govern swaps and make the high-level decisions. These governing bodies are responsible for approving the swap agreements and their basic terms such as length of agreement, amount, acceptable collateral, duration of draws, interest rates, exchange rates, and any special terms (see Key Design Decision No. 5, Administration). The governing body also often approves any extensions of the arrangements, amendments to terms, approval of supplemental agreements, or change of form of agreement such as conversion to a standing agreement. For example, the chair of the FOMC must preapprove all amendments to existing swap arrangements (FOMC 2021a). The governing bodies are also responsible for adopting repo frameworks and approving the agreements made under them. These bodies also approve any amendments or extensions to the frameworks, as when in July 2021, the FOMC changed the FIMA Repo Facility from a temporary facility to a standing facility (Fed 2021b), or when the ECB Governing Council extended its EUREP facility until 2024 (ECB 2022a).

We also observe that the highest level of lending central banks authorizes draws under a swap agreement, but with some nuances. For example, as set forth in the Authorization for Foreign Currency Operations, the chair of the FOMC must preapprove all draws under the standing or temporary dollar-liquidity agreements (FOMC 2021a). The entire FOMC must preapprove draws under the reciprocal NAFA with the BoC and Banxico (see US–Mexico 1994) and under any foreign currency liquidity agreement (FOMC 2021a). In this area, authorized parties may delegate authority to streamline processing. For the reciprocal NAFA and standing foreign currency swaps, if the FOMC is not available to act, then the three-member Foreign Currency Subcommittee[13] can act in its place (FOMC 2021a).

[13] The Foreign Currency Subcommittee consists of the chair and vice chair of the committee and the vice chair of the Board of Governors (or another Board member designated by the chair as an alternate if a member of

In some limited circumstances, approval by the government in addition to the central bank was required regarding the establishment, amendment, or extension of swap arrangements.[14] We saw this in our cases involving the PBOC and in those involving the RBI under the SAARC framework, where the RBI is the sole funder and lending bank (RBI 2012). All subsequent extensions and amendments to the SAARC framework also require Union Cabinet approval (RBI 2012).

Some governing bodies delegate some of their authority regarding swap and repo lines to lower-level bodies and often adopt guidelines that these administrative bodies rely on as guidance.

Governing bodies may delegate the authority to handle the administration of swaps and repo lines to lower-level bodies or departments to streamline processes. An example of this is the Fed's Authorization for Foreign Currency Operations and related Foreign Currency Directive. The Authorization sets out the broad delegation of administrative authority to the Federal Reserve Bank of New York (FRBNY, the selected Reserve Bank) and the general parameters of foreign currency operations that it should follow in conducting such operations on the FOMC's behalf. The Directive specifically directs the FRBNY to manage the enumerated swap lines in accordance with the Authorization. Both documents are internal procedural mechanisms that the FOMC adopted to facilitate its work, which it can amend from time to time, as it did on March 19, 2020, to provide for operation of the temporary dollar-liquidity swap lines that it authorized (FOMC 2020b; FOMC 2020c). In bifurcating these administrative guidelines, the FOMC can efficiently modify the Directive as swap lines change without the need to disturb the broader guidelines embodied in the Authorization. The Directive was again amended on January 25, 2022, to remove the enumerated temporary US dollar-liquidity swap lines (FOMC 2022). Administration of the FIMA Repo Facility is also delegated to the FRBNY (FOMC 2022).

The ECB's Governing Council[15] is the body that makes the high-level decisions regarding currency swap agreements. The ECB has adopted a "main framework within which it uses strict criteria to assess the conditions under which to grant swap and [bilateral] repo lines to non-euro area central banks" (ECB 2022c).[16] The ECB has not disclosed this document, which the Governing Council uses to "assesses incoming requests for liquidity lines on a case-

the Board serving on the subcommittee is unavailable, and the alternate of the vice chair of the committee if the vice chair of the committee is unavailable) (FOMC 2021a).

[14] We also saw approval by the US Treasury of several swap lines in our US–Mexico 1982 and US–Mexico 1994 cases. However, these were swap lines extended by the Treasury under the ESF separately from any Fed swap lines. The Treasury swap lines were also administered by the FRBNY as the fiscal agent of the government (Swaminathan 2023b; Swaminathan and Wiggins 2023).

[15] The ECB's Governing Council consists of 25 total members—six from the Executive Board and the governors of the 19 national central banks (Lawson 2020).

[16] *The ECB's Governing Council determined principles for liquidity assistance to non–euro area EU countries on October 20, 2008* (ECB 2008c). *These principles are summarized in the Agreement on Emergency Liquidity Assistance (ELA), published on November 9, 2020. This emergency assistance document will be reviewed in 2023* (ECB 2020e). *See also Key Design Decision No. 7, Eligible Institutions.*

by-case basis" (ECB 2022c). However, it has disclosed some of the criteria it considers; see Key Design Decision No. 7, Eligible Institutions.

In practice, the ECB's Executive Board reviews the proposed terms of a request for a swap arrangement from a non–euro area country, such as Hungary, against the framework and submits an analysis and recommendation to the Governing Council, which makes the final decision. In October 2008, the Governing Council decided to accept the recommendation of the Executive Board to grant Hungary a bilateral repo line (requiring it to deposit euro-denominated securities as collateral) instead of the swap line it had requested (Gupta 2023a).

The Governing Council also approved the EUREP and its basic terms (eligibility, interest rate, and collateral) (Lawson 2020). Approval of the applications under EUREP are also made by the Governing Council (ECB 2022b).

Governance of the multilateral networks—SAARC, ASA, and CMIM—is also vested in the highest level of the authorized entity. For example, the RBI makes the decisions under the SAARC. Under the ASA and CMIM, the finance ministers and governors of the central banks from the 13 participating countries make the decisions. Originally, the CMIM agreement established the Ministerial Level Decision Making Body (MLDMB)—initially made up of only the ASEAN + 3 finance ministers—as the body responsible for voting on CMIM governance decisions (for example, total size and contribution amounts) through consensus approval (AMRO n.d.; APT 2010). In July 2014, the CMIM was amended to include the central bank governors in the MLDMB and thus in governance decisions (Han 2022). Deputy-level representatives of the participating central banks and finance ministries vote on decisions related to swap draws, renewals, and cases of default (AMRO n.d.). As discussed in Key Design Decision No. 5, Administration, the MLDMB has also adopted certain administrative practices, including designating an Agent Bank, to facilitate efficient administration of the network.

> **Governance (B): Typically, central banks disclose the existence of swap arrangements and their key terms, but disclosure of the swap agreements and draws varies by bank. The Fed makes weekly aggregate disclosures under its FIMA Repo Facility, while the ECB does not disclose EUREP usage.**

Although we were not able to identify all applicable disclosure requirements, central banks generally contemporaneously publicly announced entering into a swap agreement, identifying the counterparty, maximum amount, collateral, and duration of the agreement. Parties may do this to signal the availability of funds or because of a legal requirement. In all but three of the 67 swap and repo arrangements in our cases, Lenders announced the swap agreements on signing (exceptions include Eurozone–Sweden 2007; China–Argentina 2014; and China–Mongolia 2011, which was delayed) and in all of our cases, the Lenders and Borrowers later disclosed the arrangements in their annual reports. The PBOC also disclosed the dates and sizes of renewals and amendments of its swap lines in its renminbi internationalization reports (Arnold 2023c). The ECB and Riksbank purposely delayed announcing their swap arrangement in 2007 (ECB 2007); however, both parties later disclosed the existence of the agreement in their annual reports issued in 2008 and announced the first draws several years later (Gupta 2023d). The Bank of Mongolia (BOM) did not disclose that it had signed a swap agreement with the PBOC, but it disclosed it in its annual report, and both parties issued press releases when they extended the agreement (Arnold 2023c).

One of our cases describes an interesting situation that led to increased disclosure. The United States' significant aid to Mexico through swaps in 1994 was controversial in Congress. Since it appeared Congress would not approve an aid package to Mexico, President Clinton and his secretary of the Treasury relied on the Treasury's Exchange Stabilization Fund (ESF) and the Fed to provide USD 20 billion in medium-term assistance (Lustig 1997). Congress answered with a bill that required the president to make additional disclosures regarding use of the ESF (Swaminathan and Wiggins 2023).

Several of the central banks participating in the Major Bank Swap Network have disclosed their actual swap agreements (the ECB, Fed, BOJ), but the Fed has not disclosed swap agreements with counterparties other than those in the network. The ECB has disclosed some of its swap agreements and some of the Executive Board memoranda relied on by the Council in considering whether to enter into a swap or repo.

The Fed does not disclose information regarding participants that have applied for or been approved for a repo under the FIMA Repo Facility (Fed 2020e). However, repo counterparties can disclose their access to and use of the facility if they so choose to (see Key Design Decision No. 6, Communication) (Fed 2020e). The central banks of Chile, Colombia, Ghana, Indonesia, Sri Lanka, and Hong Kong announced in 2020 that they had signed agreements with the Fed to access the facility, and Hong Kong publicized its use of the facility (Kelly 2023). Similarly, the ECB has said that it does not disclose the names of any repo counterparties approved under the EUREP (ECB 2022d), but we know that the ECB has approved a number of countries to use the facility because two counterparties have chosen to disclose this fact. In 2021, the Central Bank of the Republic of Kosovo disclosed in its 2020 annual report that it had signed a repo agreement under the facility for EUR 100 million (CBK 2021). In 2022, the Central Bank of Montenegro announced that it had also gained access to the facility for EUR 250 million (CBCG 2022). Neither country has said publicly whether it ever drew on those lines.

Generally, most central banks do not disclose information regarding possible counterparties that they have considered for swap or repo arrangements but not entered into an arrangement with, most likely out of concern for stigmatizing the unsuccessful candidate.[17] In a similar vein, many borrowing central banks make public the names of their downstream borrowers of foreign currency only years later (Bahaj and Reis 2021). The SNB does not disclose which banks borrowed US dollars under its downstream lending operations funded through its swap line with the Fed (Ostroff and Patrick 2022). Central bank disclosure of draws under individual swap or repo arrangements varies. The Fed discloses the aggregate amounts of swaps and FIMA repos outstanding in its H.4.1 report—the Fed's weekly

[17] Candidates may be identified in meeting transcripts that are later released.

snapshot of its balance sheet. It also regularly discloses swap activity by individual foreign central bank in releases on the FRBNY's website (Fed 2023a).

The ECB publishes weekly data on the aggregate daily amount of liquidity provided across all central bank liquidity lines (swaps and repos) established under its main framework and the EUREP (ECB 2022d). Additionally, on January 17, 2023, the ECB said that EUREP was "being used in response to the uncertain environment caused by the Russian invasion of Ukraine" (ECB 2022c).

The PBOC disclosed the existence of swap lines but did not provide additional information in its annual reports. It disclosed the dates and sizes of renewal, as well as amendments of its swap lines in its renminbi internationalization reports (Arnold 2023c). Its counterparty borrowing central banks sometimes disclosed amounts drawn, but to varying degrees of transparency (for example, sometimes via enumerating individual draws and durations, but usually by stock amount as line items on their balance sheets) (Arnold 2023a; Arnold 2023b; Arnold 2023c). The BOM did not disclose that it had signed a swap agreement with the PBOC, but it disclosed it in its annual report, and both parties issued press releases when they extended the agreement (Arnold 2023c).

5. Administration (A): Swap lines are generally simple arrangements that parties can deploy fairly quickly.

A swap framework agreement usually runs less than 10 pages (see, for example, ECB 2007; ECB and SNB 2003; FRBNY 2010). Once the framework agreement is in place specifying the key terms (for example, amount authorized, exchange rate, interest rate, collateral term), eligible parties can draw funds fairly swiftly; speed is also supported by the simplicity of the collateral. This makes swaps a useful tool in urgent situations. After the September 11, 2001, terrorist attacks, when normal transmission channels were disrupted, the Fed entered into swap agreements with the ECB, BoE, and BoC to provide US dollars. Within one day of the attack, on September 12, the ECB began drawing on the swap line (FRBNY 2001). The ECB borrowed more than USD 19 billion, which it lent to its banks; the first draws occurred prior to publication of the press release announcing the swap. Repayment occurred within a week, and the line was terminated a month later when conditions had returned to normal (French 2023d).

Administration (B): For efficiency, the Lender governing body often delegates the administration of the swap or repo arrangement to a lower-level body while maintaining some oversight.

The high-level governing bodies that establish swap or repo arrangements often delegate some of their authority and technical details to administrative agents that operate according to guidance the governing body provides. This guidance can promote efficiency and provide some assurance that the same terms are being applied across all potential participants.

For example, the Fed delegates authority to the FRBNY under its Authorization for Foreign Currency Operations and the related Foreign Currency Directive that the FOMC adopted (see Key Design Decision No. 4, Governance). The FRBNY thus establishes Borrower correspondent accounts and processes requests for draws as the FOMC's agent. Still, the final approval of draws has to be made by either the chair (for the US dollar swap lines) or the FOMC (for the NAFA or foreign currency swap lines) (FOMC 2021a). For the Fed's dollar swap lines, the Chair may preapprove a schedule of potential swap drawings and delegate to the FRBNY the approval of individual drawings under that schedule (FOMC 2021a) (see Key Design Decision No. 9, Process for Utilizing the Swap Agreement).

The FRBNY also administers the FIMA Repo Facility operations pursuant to a Standing FIMA Repurchase Agreement Resolution adopted by the FOMC (FOMC 2021b). The FRBNY services deposit and custody accounts for more than 200 foreign official and international institutions, most of which are central banks and other international monetary authorities. FIMA account holders are eligible to apply to the FOMC to use the FIMA Repo Facility (Kelly 2023). The FOMC also authorized its Foreign Currency Subcommittee to change the FIMA Repo Facility's offering rate, term, and counterparty limits provided the Subcommittee keeps the Committee informed of any such changes (FOMC 2020f).

The ECB has also adopted a memorandum providing guidance for reviewing requests for swap agreements with non–euro area countries. The Executive Board applies the guidance and provides an analysis and recommendation to the Governing Council (ECB 2008a; ECB 2022c).

We do not have much detail about which internal entities or departments of other central banks administer swap arrangements and draws, but Key Design Decision No. 9, Process for Utilizing the Swap Agreement, describes the several ways that various central banks process draws.

The ASA and CMIM both have relatively decentralized administrative processes, coordinated by a rotating member (two for the CMIM), reflective of their structure as commitment pools. To facilitate administrative functions, the ASA appointed a participating central bank as the standing Agent Bank, serving on a two-year rotational basis based on the alphabetical order of the country (ASEAN 2005; Bank Negara Malaysia 2000). The Agent Bank administered the drawing process—initiated by a member's submitting a request to the Agent Bank. Once received, the Agent Bank communicated the request to other member central banks, which indicated their participation. The Agent Bank then coordinated the draw and handled any renewal requests. The Agent Bank was also responsible for any administrative expenses (ASEAN 2005).

Similarly, CMIM members elect two representatives to administer the network: one from an ASEAN country and one from a Plus Three country. The representatives coordinate CMIM discussions and negotiations over CMIM activation (AMRO n.d.). Activation-related decisions (drawings, renewals, defaults) are approved through a two-thirds majority vote of the CMIM's Executive Level Decision Making Body (ELDMB), composed of the deputy officials of the member countries' central banks and finance ministries. Some commentors have expressed concerns that such administrative decentralization impedes the CMIM's ability to respond to crises efficiently and quickly (Han 2022; Hoffner 2023b). Despite the occurrence of several crisis situations since its inception, including the COVID-19 pandemic, the facility remains untapped, while some members have utilized other sources of foreign exchange liquidity (such as through the Fed's swap and FIMA Repo facilities) (Negus 2020). There is uncertainty around its operational readiness and its lack of "rapid response procedures to handle a fast-developing financial emergency" (Menon and Hill 2014, 10). Its processes for approving draws have been characterized as "slow, cumbersome and politically complex, producing widespread skepticism in its ability to provide assistance" (Drysdale, Basri, and Triggs 2020).

6. **Communication: Lending and borrowing central banks typically announce when they put swap or repo arrangements in place, create downstream lending programs, and extend the deadlines of these arrangements. Less often, they announce when Borrowers and downstream financial institutions draw on funds, or when a swap arrangement terminates.**

Central banks typically announce that they have entered into swap and repo arrangements to address a liquidity constraint or market distress. Announcements of swaps can bolster the impact of the announcement of a downstream lending program by the borrowing central bank, indicating its commitment to take direct and immediate action to address the problem. The memo of the ECB Executive Board analyzing Hungary's request for a swap arrangement indicated that the MNB "would like to make public the existence of the agreement/facility: it expects the positive signaling effect of this announcement to contribute to the credibility of the newly created MNB's [downstream] foreign exchange swap facility" (ECB 2008a, 2). In a similar manner, in the press releases issued by the SNB, MNB, and NBP in 2008 regarding their bilateral swap whereby the SNB was to provide Swiss francs to the MNB and NBP, both borrowing central banks also announced downstream lending of Swiss francs to their domestic banks. The press releases, often issued jointly, discussed the downstream foreign exchange swaps and the central bank swaps together. Borrowing central banks also may announce their willingness to draw on a swap line to offer future downstream lending if necessary, as the ECB did with respect to pounds sterling in the prelude to Brexit (ECB 2019).

When two parties have a standing swap arrangement, they typically make a separate announcement when they have activated it. Activation usually means that some action has been taken to move the swap closer to use. It can be a useful signal to markets.[18] In 2019, the ECB and BoE activated their standing swap leading up to the UK's exit from the European Union "for the possible provision of euro to UK banks" (ECB 2019). Concurrently, the BoE announced a new Liquidity Facility in Euros (LiFE) under which it would lend euros to UK banks on a weekly basis. The LiFE program would be "facilitated by the activation of the standing swap line" (BoE 2019). The ECB also stated in its press release that it "would stand ready to lend pound sterling to euro area banks, if a need arises" (ECB 2019).

In March 2020, repeating a practice that had been relied on during the GFC and that had been received as very effective, the six major central banks of the Major Bank Swap Network jointly announced the activation of their agreements in cross-linked press releases with almost identical wording. (See BoC 2020; BoE 2020; BOJ 2020; ECB 2020a; Fed 2020a; SNB

[18] Per correspondence with Saleem Bahaj, research manager at the Bank of England.

2020.) The Fed's 2020 press release described swap measures as "coordinated central bank action to enhance the provision of US dollar liquidity" (Fed 2020a). The six central banks announced that they would lower the rate for US dollar swap arrangements by 25 basis points (bps), introduce a 84-day maturity term, and increase the frequency of existing one-week maturity loan operations from weekly to daily (ECB 2020b; Fed 2020a).[19]

In practice, if not in wording, these releases served as a "reactivation" of these lines, as they announced that the central banks stood ready to lend under them. Collectively, they were also a strong statement of cooperation among the central banks to forestall interruption in the flow of US dollars (Fed 2011; Fed 2020a).

There appeared to be a small difference between the BoE's and ECB's announcements of their 2010 swap line. The ECB said the swap would provide pounds to the ECB if needed. The ECB announcements, and those of the CBIR regarding the related ECB–CBIR line, stated that the ECB–BoE line could be used only to provide pounds to the CBIR for loans to Irish banks (Wiggins 2023). It's not clear why the descriptions differed, but the Borrower determines the use of the borrowed funds, including any restrictions on their use, so it was most appropriate for the ECB to announce this if it desired.

Differences in communication from parties to a swap can create confusion. For example, in 2014, the Central Bank of Argentina (BCRA) issued a press release that described its revised swap line agreement with the PBOC as a significant improvement on the original 2009 agreement. Later that same day, the BCRA issued a second release that struck wording from the first and further described the swap line as a demonstration of BCRA's and PBOC's "close relationship" and the two countries' "comprehensive strategic partnership" (BCRA 2014). It also described the renminbi as "a very attractive investment currency" (BCRA 2014).[20] Meanwhile, the BCRA did not withdraw the original release, creating uncertainty about how the BRCA wanted the facility to be perceived. The PBOC did not announce the line until 2015.

Swap arrangements can also invite political controversy. Legislators in the US and public commentors in China have criticized extending swap lines to some countries (Steil, Della Rocca, and Walker 2021). One scholar has called PBOC swap lines a form of "financial statecraft" and has said that the PBOC's extension of swap lines may be used as leverage to extract political concessions (McDowell 2019, 135-137). In 2017, the PBOC froze negotiations with Mongolia including regarding a renewal of its swap with the BOM because of the Mongolian government permitted a visit from the Dalai Lama (Panda 2016). Only after the government made certain consoling statements and agreements to China was the swap renewed (Arnold 2023c; BOM 2021; Zhang 2017).

7. Eligible Institutions: Most major lending central banks limit the countries they have swaps with to those with which they have a close geographic, political, and economic relationship, particularly as it relates to financial stability. The PBOC,

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[19] At the time of the first joint announcement, all partner central banks except the BoC offered US dollar-liquidity operations (Fed 2020a). A few days later, the BoC announced its intent to offer a USD Term Repo Facility if needed utilizing dollars from its swap with the Fed.

[20] The quoted texts are the authors' translations of the Spanish-language press release.

on the other hand, has entered into a large number of swaps, many with emerging market economies, to encourage trade and disseminate the renminbi.

Lending central banks do not typically disclose the criteria that they use when deciding whether to extend swap lines to another central bank (Bahaj and Reis 2021). They also do not typically disclose that they have rejected another central bank's request for a swap agreement, out of concerns about confidentiality and stigma (Sahasrabuddhe 2019).

Sovereign credit risk is an important consideration for most lending central banks. The second- and third-largest Lenders (in numbers of swap lines), the Fed and ECB, are selective in choosing swap counterparties, reflecting their focus on swap lines as financial stability tools and their concerns about credit risk. The ECB and Fed both use repo arrangements to extend liquidity to a much broader range of counterparties.

The Fed's Counterparties

During the GFC, the Fed rolled out swap lines to 14 central banks in four stages. It set up lines with the ECB and SNB in December 2007 to shore up funding for European banks and prevent spillovers into the US (FOMC 2007a). It later established lines with the UK, Japan, and Canada after the collapse of Lehman Brothers in mid-September 2008, and with five smaller advanced economies—Australia, Denmark, New Zealand, Norway, and Sweden—later in the month and in October (Allen and Moessner 2010). Fed officials internally described the final four countries to receive swap lines—Brazil, Korea, Mexico, Singapore—as emerging market economies (FOMC 2008). The FOMC discussed at length whether or not to extend these lines and considered alternatives, such as repos, to provide additional security, and in the case of Mexico, enlarging the existing NAFA swap line (FOMC 2008). In an October 2008 meeting, FOMC members agreed that, although emerging market economies, these countries had strong macroeconomic fundamentals and the credit risk to the Fed would be minimal. However, some members argued that the approval of these lines would lead to many requests from other central banks, which would require individual consideration in the absence of a bright-line test. The question, left unresolved, became where to draw the line (FOMC 2008).

Nathan Sheets, a former director of the Fed's International Finance Division, described three criteria that the Fed staff used in making its recommendations:

- Is it a large, systemically important economy?
- Is the economy one in which the country's policies have been strong and it appears that it is largely being influenced by contagion? and
- Is it a country for which we believe that the swap line might actually make a difference? (FOMC 2008)

Fed officials also discussed the likelihood that these four countries would request IMF assistance but agreed it was unlikely. Fed officials viewed the IMF as better suited to providing financial support to smaller countries (FOMC 2008). The Fed also recognized that several emerging market economies were large holders of US Treasuries; without access to US dollar liquidity, there was a risk they would liquidate their holdings (Sheets 2019). A US Government Accountability Office audit revealed that the Fed had considered other criteria: whether recipient countries were large trading partners, held high levels of foreign reserves, or had financial linkages with US banks (Sahasrabuddhe 2019).

Sahasrabuddhe (2019) argues that the Fed also had geopolitical motives. Brazil had shown promise in financial liberalizing and had an influential role in the Group of 20; Korea had rising political influence in Asia, growing economic ties to the US, and large foreign exchange reserves; Mexico was a neighbor with a close economic relationship to the US, and its stability was a national security concern; and Singapore was an important global financial center, and also demonstrated strong economic fundamentals (Sahasrabuddhe 2019). Several countries said that the Fed had rejected their requests for swaps during the GFC: Chile, the Dominican Republic, Iceland, India, Indonesia, Peru, and Turkey (Sahasrabuddhe 2019).

Unlike the PBOC, the Fed has not expanded its swap lines beyond its 14 GFC counterparties. The Fed and the central banks of Canada, the euro area, Japan, Switzerland, and the UK agreed to convert their reciprocal swap arrangements into an uncapped standing multilateral network in 2013, the Major Bank Swap Network. During the COVID-19 crisis, the parties activated the network, and the Fed again extended temporary US dollar–liquidity lines to the other nine central banks that had lines during the GFC (Fed 2020c). Aizenman, Ito, and Pasricha (2021) find large trade ties with the US to be the most significant factor explaining the Fed's decision to reinstate swaps with these nine countries. The authors also find that during the GFC, the dominant factor for entering into swaps with the emerging market economies was their financial ties with the US. For both crises, they find that the existence of formal military alliances was also a determinant.

In the COVID-19 pandemic, several countries without Fed swap lines began to sell their Treasuries (Choi et al. 2022; Singh 2023). The Fed responded by establishing the FIMA Repo Facility to provide dollars against Treasury collateral, a development that was responsive to arguments from emerging market economies for greater expansion of US swap lines (Aizenman, Ito, and Pasricha 2021; Fed 2020d; Fed 2020e). About 170 central banks and foreign entities were eligible to apply and 30 ultimately enrolled (Smialek 2023). The Fed's swap lines and repo facilities combined effectively cover 80% of foreign official holdings of Treasuries. To prevent stigma, the Fed does not disclose the identities of FIMA participants (Singh 2023).

The ECB as Regional Lender of Last Resort

Since its inception in 1998, the ECB has acted as a "regional lender of last resort," providing euro liquidity to European countries (Albrizio et al. 2022). Like the Fed, the ECB restricts swap line eligibility to a narrow set of counterparties. Prior to the GFC, it had some precautionary swap agreements with other G-10 central banks such as the BoE and the SNB, and it later came to participate in the Major Bank Swap Network. Also, during the GFC, it entered into bilateral swap lines with Sweden and Denmark and repos with Hungary and

Poland (ECB 2022c). Spielberger (2023) points out that the ECB made a "clear geographic distinction between the Nordic and Eastern European central banks" (881). The swap lines to Sweden and Denmark were considered equal to the ECB's lines with G-10 countries (for example, the UK and Switzerland) and quickly processed, while the liquidity lines to the Eastern European countries were debated and resulted in repo agreements rather than swap lines to better protect the ECB's balance sheet (Spielberger 2023). See also Key Design Decision No. 13, Balance Sheet Protection.

At the time it considered swaps to Hungary and Poland, the ECB had not adopted guidelines as to how to determine whether to provide liquidity lines to non–euro area central banks; guidelines were adopted a short time later, but not disclosed (Spielberger 2023). The ECB stated in 2014 that it looks to the following criteria in determining whether to provide a swap or repo line:

- The existence of exceptional circumstances characterized by significant actual or potential euro liquidity needs as a result of serious market disfunctions;
- The systemic relevance for the euro area of the country requesting the swap line and the exposures of the euro area banking sector and financial markets to that country;
- The presence of sound economic fundamentals;
- The financial risk for the Eurosystem; and
- The consistency with any parallel support provided by the IMF. (ECB 2014)

As of 2022, in addition to the standing Major Bank Swap Network, the ECB has swap lines with the central banks of China, Denmark, Poland, and Sweden (ECB 2022c). As shown in Figure 7, it also has bilateral repos with six counterparties.

In June 2020, the ECB established the EUREP facility to provide euro liquidity—secured by euro-denominated sovereign debt—to non–euro area central banks without access to swap lines or repos under its main framework (ECB 2020c; ECB 2022c). Similar to the Fed's FIMA Repo Facility, the EUREP offers an additional tool to extend euro liquidity beyond the otherwise narrow eligibility criteria of swaps and bilateral repos. A broad range of non–euro area central banks are eligible to apply, with decisions being made by the Governing Council. The ECB did not disclose additional restrictions on eligibility (ECB 2022c).

The Role of Close Connections

For other lending central banks, our cases seem to indicate that which countries get swaps is greatly influenced by geographic proximity and economic ties between the Lender and Borrower countries. Sixteen of our 23 cases include swap and repo arrangements between neighboring or geographically close countries; the exceptions are PBOC and Fed swaps, as discussed. Regionalism alone may not fully explain eligibility in each swap arrangement but may point to other explanatory variables such as financial integration, trade, and various political factors, similar to those cited in the implementing documents of the ASA, CMIM, and

SAARC, swap arrangements that emerged out of existing regional cooperation frameworks. In a somewhat similar circumstance, in 2003, the central banks of Denmark, Finland, Iceland, Norway, and Sweden signed a memorandum of understanding establishing general principles of cooperation during financial crises (Suomen Pankki 2003). A 2008 Moody's Investors Service report recommended Iceland expand its financial safety net by soliciting support from the countries with which it already had a cooperative agreement and a history of political and economic cooperation owing to the region's close geographic, cultural, and commercial ties (Hoffner 2023c; Moody's 2008). The result was the Scandinavia–Iceland swap lines. Also, in 2010, at the time the BoE provided a pound sterling swap line to the ECB for the benefit of Ireland, much of UK government commentary emphasized the significant financial connections between the UK and Ireland.

8. **Size: Lender central banks typically set the size of swap lines in some relation to the size of the economy of the Borrower, whereas repo arrangements are usually limited only by the quantity of eligible collateral that the Borrower can provide. The only unlimited swap arrangements we came across were in the Major Bank Swap Network.**

Most lending central banks set limits on the amount a Borrower can have outstanding on a swap line at any time. In several cases, the Fed has also set limits on individual draws. The only unlimited swap arrangements we came across were in the Major Bank Swap Network, which includes the Fed and five other major central banks, which began with limits. Repo lines are typically limited only by the amount of eligible collateral available to the Borrower. The Fed did not initially set a size limit on the usage of the FIMA Repo Facility but added a counterparty limit of USD 60 billion when it made the facility permanent in 2021 (FOMC 2021b).

In setting the size of a swap line, policymakers face a trade-off. Larger swap lines can provide more liquidity in foreign jurisdictions during a crisis, but they also expose the Lender to greater balance sheet risk. Lenders can manage that risk by approving each swap draw and limiting their size.

Central banks disclose very little information about how they determine the appropriate size of a swap line. In our survey, line sizes often reflected the size of a Borrower's economy. Figure 9 shows the size of the Fed's swap lines scaled by the borrowing country's reserves and GDP. In addition to economic variables, political factors may also influence the size limits on swaps. The Scandinavian swap lines to Iceland, the CMIM arrangement, and the Fed swap lines offer some insight. In the Scandinavia–Iceland swaps, the Central Bank of Iceland requested swap lines of EUR 1 billion to EUR 2 billion from several central banks. Ultimately, the three Scandinavian central banks were the only ones to agree, and each established with Iceland a bilateral swap line of EUR 500 million, for a total of EUR 1.5 billion (SIC 2010, 1:173–76). Although the economies and financial resources of the Scandinavian countries differed, they nonetheless established equivalent swaps with Iceland.

In another example, the Fed maintained swap lines with 14 central banks during the GFC and COVID-19 pandemic. Its lines are unlimited with its five counterparties in the Major Bank

Swap Network, although the line with the Bank of Canada was limited before 2014. To avoid the appearance of ranking its counterparties, according to officials at the Fed and the FRBNY, the Fed divided the other nine counterparties into two groups by size. Six central banks received lines of USD 30 billion and three received lines of USD 15 billion during the GFC. The Fed doubled the size of those lines in the COVID-19 crisis to reflect the counterparties' economic growth since the GFC (Fed 2020c). As seen in Figure 9, the sizes of these two groupings do not neatly reflect the economic variables of each country. Four countries—Brazil, Korea, Mexico, Singapore—the "emerging market countries," received smaller swaps relative to their reserves, on average, than the five "smaller advanced countries" of Australia, Denmark, New Zealand, Norway, and Sweden (FOMC 2008, 9–10).

Another US swap, one of the US swap lines extended to Mexico in 1994, included a term regarding size not usually seen in a swap line. Once a drawing under the Special Swap Line was repaid, the authorized size of the line was reduced pari passu by the amount of the draw, each use shrinking the available amount. Most swap arrangements are revolving in nature up to the maximum authorized amount (FOMC 1995; Swaminathan and Wiggins 2023).

The ECB has not disclosed much information about how it chooses which counterparties to enter into swaps with, but its Governing Council determined principles for liquidity assistance to non–euro area EU countries on October 20, 2008 (ECB 2008c). These principles are summarized in the Agreement on Emergency Liquidity Assistance (ELA), published on November 9, 2020. The Executive Board refers to those criteria when reviewing potential candidates for swaps or repos and making recommendations to the Council, including with respect to size of the liquidity facility (Gupta 2023b).

Lending central banks may also set limits on their contributions to multilateral swap arrangements. In the ASA, the 10 ASEAN members were divided into two tiers. Tier 1 countries each contributed 15% of total commitments. Tier 2 countries contributed less than that, with the amounts varying depending on each member's ability to pay (ASEAN 2005; Rana 2002). Similarly, the CMIM divides countries into tiers based on their ability to pay and the size of their economies. China and Japan each contribute 32% of total funding, Korea commits 16%, and the remaining 10 CMIM members contribute the remainder. Han (2022) notes that Japan and China would contribute more than 80% of CMIM funds if their contributions reflected the relative size of their economies or reserves (Han 2022). The two countries underweighted their contributions to avoid the appearance of uneven influence (Ciorciari 2011).

The CMIM also sets limits on the maximum size of a swap allocation (total outstanding draws), based on a multiple of a member's contribution. However, that multiple is not the same for every country. The multiple is just 0.5 for China and Japan. It is 5.0 for the smallest five countries, to ensure they have access to adequate liquidity in a crisis (Han 2022).

Small Swap Lines

Some of the swaps we studied were small and more valuable as symbols of financial cooperation than as tools for financial stability. For example, in 1977 the five original parties to the ASEAN Swap Arrangement made US dollar funding commitments of just USD 20 million each, or USD 100 million total. They doubled those commitments in 1978. Due to its small size, the ASA could provide only modest support to members in need of foreign liquidity and was not used during any major financial crisis (Chandrasekhar 2021; Henning 2002, 14).

The three Scandinavian swap lines to Iceland (EUR 1.5 billion in total) were small relative to Iceland's USD 170 billion financial system. The governor of the CBI said that the size of the swaps was less important than the symbol of solidarity and attempted to secure swaps of similar size with other central banks (SIC 2010, 179). In rejecting the CBI's swap request, then–BoE Governor Mervyn King said that "the amount of money is very small relative to the potential need for funds should a problem arise with one or more of your banks" and that "the swap might look rather like a political gesture rather than a credible financial strategy" (SIC 2010, 173). The Fed also rejected the CBI's request. Internally, a Fed official cited Iceland as an example of a country that did not meet one of his criteria for a Fed swap line: "that the swap line might actually make a difference" (FOMC 2008, 33) (see Key Design Decision No. 7, Eligible Institutions).

Swap Line Increases

In 19 of 67 swap and repo arrangements in our survey, Lenders authorized at least one increase, and in many, more than one. Lenders are more likely to increase swap lines in prolonged crises than in shorter stresses. During the GFC, the Fed increased eight of its swap lines several times (prior to making four lines unlimited). When it reinstated nine temporary US dollar–liquidity swap lines during the COVID-19 crisis, the Fed authorized amounts under these lines that were twice the highest amounts authorized to these counterparties during the GFC. (See Figure 9.) The Fed said, "Their limits were raised to reflect that the global economy has grown since 2008, both in real and in nominal terms, and therefore larger amounts of funding might be needed to address levels of stress in dollar funding markets" (Fed 2020c).

Usage

The most heavily used swap lines have been the Fed's US dollar swap lines, particularly during the GFC and the COVID-19 pandemic. During the GFC, amounts outstanding under Fed swaps peaked at USD 583 billion and during the COVID-19 crisis at USD 449 billion (Choi et al. 2022; FRBNY n.d.a). As seen in Figure 9, the ECB was the dominant user during the GFC, with a peak outstanding of USD 314 billion, and the BOJ was the dominant user during COVID-19, with a peak outstanding of USD 225.8 billion. Though the BOJ, BoC, BoE, ECB, and SNB have all had standing swap lines in place since 2013, usage has largely been confined to crises. Multiple academic studies point to the effectiveness of the Fed's heavily used swap lines during the GFC and the COVID-19 pandemic in reducing strains in global US dollar funding markets (Bahaj and Reis 2020; Choi et al. 2022).

However, of our 67 surveyed swap and repo arrangements, 14 were unused. With respect to the Fed's swap lines, a number of lines went unused after the September 11 terrorist attacks, during the GFC, the SDC, and the COVID-19 pandemic. Nevertheless, government officials and scholars have commented positively on the benefits of swap lines as liquidity backstop facilities, supports of financial stability, and evidence of central bank cooperation, arguably validating such arrangements even when usage is absent or minimal (Bordo, Humpage, and Schwartz 2014; Fed 2020g; McCauley and Schenk 2020).

Figure 9: Size—Authorized Amounts

Case	Swap/Repo Arrangement	Peak Usage (or Highest Reported)	Original Amount Authorized	Maximum Amount Authorized		
		USD Billions	USD Billions	USD Billions	% of Borrower's Reserves	% of Borrower's GDP
ASA 1977	ASA 1977	No info	0.1	2.0	N/A	N/A
ASEAN +3 CMIM 2010	ASEAN +3 CMIM 2010	No usage	120.0	240.0	N/A	N/A
China-Argentina 2014	China-Argentina 2014	20.5	11.3	22.7	50.7%	No info
China-Hong Kong 2009	China-Hong Kong 2009	8.7	29.2	119.1	28.1%	No info
China-Mongolia 2011	China-Mongolia 2011	1.9	0.8	2.3	50.6%	17.3%
Eurozone-Denmark 2008	Eurozone-Denmark 2008	4.2	16.8	16.8	88.8%	10.6%
Eurozone-EUREP 2020	Eurozone-EUREP 2020	No info	No info	No info	N/A	N/A
Eurozone-Hungary Repo 2008	Eurozone-Hungary 2008	2.5	7.0	7.0	20.7%	4.4%
Eurozone-Ireland and UK 2010	Eurozone-Ireland 2010	No info	15.6	15.6	738.0%	7.0%
	Eurozone-UK 2010	No info	15.6	15.6	20.6%	0.1%
Eurozone-Poland Repo 2008	Eurozone-Poland 2008	No info	14.0	14.0	22.5%	2.6%
Eurozone-Sweden 2007	Eurozone-Sweden 2007	3.8	14.6	14.6	No info	No info
Eurozone-UK 2019	Eurozone-UK 2013	No info	Unlimited	Unlimited	N/A	N/A
SAARC 2012	India-Bhutan 2013	0.2	0.1	0.4	No info	No info
	India-Sri Lanka 2015	1.1	0.4	1.1	15.1%	1.3%
	India-Maldives 2019	0.4	0.4	0.4	40.6%	10.7%
Scandinavia-Iceland 2008	Denmark-Iceland 2008	0.3	0.8	0.8	21.5%	4.2%
	Norway-Iceland 2008	0.3	0.8	0.8	21.5%	4.2%
	Sweden-Iceland 2008	0.2	0.8	0.8	21.5%	4.2%
Switzerland-Multiple 2008-09	Switzerland-Eurozone 2008	32.0	No info	No info	No info	No info
	Switzerland-Poland 2008	0.7	No info	No info	No info	No info
	Switzerland-Hungary 2009	0.8	No info	No info	No info	No info
US-BIS and Bundesbank 1967	US-Bundesbank 1962	No info	0.4	0.8	No info	No info
	US-BIS 1965	0.1	0.2	0.6	No info	No info
US FIMA Repo 2020	US FIMA Repo 2020	1.4	No info	No info	N/A	N/A

Case	Swap/Repo Arrangement	Peak Usage (or Highest Reported)	Original Amount Authorized	Maximum Amount Authorized		
		USD Billions	USD Billions	USD Billions	% of Borrower's Reserves	% of Borrower's GDP
US-Mexico 1982	US-Mexico 1967	0.7	0.7	0.7	119.3%	2.6%
	US-Mexico 1982 (Fed/UST)	0.6	0.9	0.9	52.0%	0.5%
	US-Mexico 1982 (UST)	0.8	1.0	1.0	56.3%	0.5%
US-Mexico 1994	US-Mexico 1994 (NAFA)	2.0	6.0	6.0	93.2%	1.1%
	US-Mexico 1994 (TSL)	No usage	6.0	6.0	93.2%	1.1%
	US-Mexico 1995 (AP)	2.5	20.0	20.0	117.3%	5.6%
US-Multiple 2001	US-Canada 2001	No usage	19.5	10.0	29.2%	1.4%
	US-Eurozone 2001	19.5	50.0	50.0	115.3%	0.8%
	US-UK 2001	No usage	30.0	30.0	63.3%	1.8%
US-Multiple 2007-09	US-Eurozone 2007	314.0	20.0	Unlimited	N/A	N/A
	US-Switzerland 2007	30.0	4.0	Unlimited	N/A	N/A
	US-Australia 2008	27.0	10.0	30.0	91.1%	2.8%
	US-Brazil 2008	No usage	30.0	30.0	15.5%	1.8%
	US-Canada 2008	No usage	10.0	30.0	68.4%	1.9%
	US-Denmark 2008	15.0	5.0	15.0	35.4%	4.2%
	US-Japan 2008	128.0	60.0	Unlimited	N/A	N/A
	US-Korea 2008	16.0	30.0	30.0	14.9%	2.9%
	US-Mexico 2008	3.0	30.0	30.0	31.5%	2.7%
	US-New Zealand 2008	No usage	15.0	15.0	135.7%	11.3%
	US-Norway 2008	9.0	5.0	15.0	29.4%	3.2%
	US-Singapore 2008	No usage	30.0	30.0	16.9%	15.5%
	US-Sweden 2008	25.0	10.0	30.0	100.9%	5.8%
	US-UK 2008	95.0	40.0	Unlimited	N/A	N/A
US-Multiple 2010-11	US-Canada 2010	No usage	30.0	Unlimited	N/A	N/A
	US-Eurozone 2010	90.0	Unlimited	Unlimited	N/A	N/A
	US-Japan 2010	21.0	Unlimited	Unlimited	N/A	N/A
	US-Switzerland 2010	0.5	Unlimited	Unlimited	N/A	N/A
	US-UK 2010	No usage	Unlimited	Unlimited	N/A	N/A
US-Multiple 2020	US-Canada 2013	No usage	Unlimited	Unlimited	N/A	N/A
	US-Eurozone 2013	145.0	Unlimited	Unlimited	N/A	N/A
	US-Japan 2013	225.8	Unlimited	Unlimited	N/A	N/A
	US-Switzerland 2013	10.9	Unlimited	Unlimited	N/A	N/A
	US-UK 2013	31.6	Unlimited	Unlimited	N/A	N/A
	US-Australia 2020	1.2	60.0	60.0	141.0%	4.5%
	US-Brazil 2020	No usage	60.0	60.0	16.9%	4.1%
	US-Denmark 2020	5.3	30.0	30.0	41.2%	8.5%
	US-Korea 2020	18.8	60.0	60.0	13.5%	3.7%
	US-Mexico 2020	6.6	60.0	60.0	30.1%	5.5%
	US-New Zealand 2020	No usage	30.0	30.0	218.5%	14.2%
	US-Norway 2020	5.4	30.0	30.0	39.9%	8.3%
	US-Singapore 2020	10.0	60.0	60.0	16.2%	17.4%
	US-Sweden 2020	No usage	60.0	60.0	103.0%	11.0%

Source: Authors' compilation.

9. **Process for Utilizing the Swap Agreement: Typically, swap agreements set out the process for utilizing the swap and only key transactional terms need to be agreed to by the parties at a draw. In most cases, a draw requires approval by the governing body or by an administrative body under preestablished guidelines set by the governing body.**

The framework swap agreement signed by the two central banks usually sets out the process for drawing funds, including size, maturity limits, duration of draws, and collateral (see FRBNY 2014 and ECB 2007). Subject to the constraints of the swap agreement, the borrowing central bank initiates the draw by requesting a specific amount of the Lender's currency and a maturity or repayment date. Once the Lender approves the request, the parties agree on the amount, the value date, the exchange rate, any swap points,[21] the maturity date, and the interest rate. Often, the swap agreement sets out the acceptable parameters for several of these terms, so they don't need further approval from the governing body of the central bank.[22]

On the transaction date, the Lender credits the borrowed currency to the Borrower's correspondent account at the lending central bank (Bahaj and Reis 2021; FRBNY n.d.c). At the same time, the borrowing central bank credits its collateral currency to the Lender's account at the Borrower's bank (Bahaj and Reis 2021). On the maturity date, the transaction is reversed. The central banks debit the respective accounts based on the same exchange rate as the initial leg of the swap, with the Lender receiving its currency back plus interest and the Borrower receiving back its currency collateral (Bahaj and Reis 2021). Central banks utilize standard industry mechanisms, such as SWIFT[23] messages, for these processes (FRBNY 2014).

Swaps vary in the speed at which the parties can activate their agreements and execute transactions.

Swaps are a relatively simple tool that can be implemented quickly. In September 2001, in response to the terrorist attacks in the US, the Fed and ECB entered into a swap agreement

[21] The SNB's swap agreement with the ECB states that the spot exchange rate would be based on the prevailing market spot rate, while the forward exchange rate would equal the spot exchange rate plus or minus forward swap points. Swap points are a discount or premium to account for the spread between the interest rates on the two currencies. In effect, the ECB paid the rate of interest on the Swiss francs it borrowed and received the interest rate on the euros it lent in exchange. Once the SNB and ECB agreed to a drawing, they agreed to the exchange rate and fees, along with other specifics (French 2023a).

[22] For example, in the agreement among the Fed, the BOJ, and other central banks during the Sovereign Debt Crisis, the interest rate for US dollar liquidity was specified as having a "minimum interest rate of the USD OIS rate plus 100 bps" (French 2023c). The SAARC framework specified that the interest rates charged under the SAARC framework would be the three-month London Interbank Offered Rate plus 200 basis points (bps) for dollar and euro swaps, and the RBI repo rate minus 200 bps for Indian rupee swaps (RBI 2012). The second rollover of any draw under a SAARC swap agreement would incur 50-bp premium to the previously charged interest rate (Gupta 2023e; RBI 2012).

[23] SWIFT stands for Society for Worldwide Interbank Financial Telecommunication and is a standard, secure method of communication between financial institutions (Wallace 2017).

and processed a draw under the agreement the same day, even prior to the press release announcing the swap line (ECB 2002; Fed 2001).

Some swaps have timing requirements that evidence the parties' recognition that swift deployment may be desirable during a crisis. For example, the agreement between the ECB and the Riksbank requires that the ECB respond to a draw request from the Riksbank on the same day that the draw request is made (Eurozone–Sweden 2007). However, central banks in the Major Bank Swap Network all require at least one-day notice between the draw request or activation and the transaction date, with the exception of Japan, which has a two-day notice due to its time zone (Bahaj and Reis 2021).

In contrast, the process for approving and settling swap draws under multilateral facilities like the ASA and the CMIM may take considerably longer because multiple decision-makers are involved (see Key Design Decision No. 5, Administration). The decentralized financing structure of the ASA and CMIM requires that at the time of a draw, each lending central bank must commit to fund swaps with the borrower (ASEAN 2005; Han 2022). In the ASA, a borrower submitting a request had to set a transaction date at least seven days after the day of the request. If one of the 10 ASEAN central banks chose to opt out of the agreement, the transaction date was pushed back an additional seven days while the remaining central banks negotiated lending amounts (ASEAN 2005). Critics have argued that these timing delays and procedural frictions may have created the perception that the arrangement is not suited for urgent liquidity needs and may explain the lack of use of both facilities (Han 2022; Hoffner 2023a; Hoffner 2023b). The SAARC framework, also a regional lending facility, differs in that the RBI is the sole lender and in charge of governance and administration for the swaps. It is perhaps because of this centralized structure that scholarship has praised the SAARC framework for timely disbursals and low conditionality (Mühlich, Fritz, and Kring 2022).

Swap draws are typically approved by the highest-level governing body of the central bank, although we observe some limited delegation for efficiency purposes.

Although administrative functions are often delegated by the governing bodies to lower-level entities, swap draw/activation requests typically still require approval from the highest governing level of the lending central bank. For the Fed, this is the chair of the FOMC, or the entire Committee as specified in the Authorization; for the ECB, it is the Governing Council. Governance-level approval of draws provides the Lender another opportunity to review and confirm not just the broad terms of the swap or repo agreement but the specific proposed terms and surrounding conditions at the time of draw. This type of approval was used by most lending central banks among our cases: DNB, ECB, Fed, Norges Bank, PBOC, and Riksbank.

Some lending central banks delegate the approval of swap draws. Typically, such approvals remain at a similarly high level. For example, if the FOMC is not available to act, then a three-member subcommittee of its members can act in its place (FOMC 2021a). Similarly, it appears that the ECB has a practice of delegating authority for implementation decisions and swap draws to one member of the Executive Board, rather than requiring the entire body to act. We saw such wording in the Executive Board memoranda to the Governing Council addressing the arrangements with the DNB, MNB, and NPB (ECB 2008a; ECB 2008b; ECB 2008c).

In a few cases, central banks have delegated the authority to approve swap draws to lower-level bodies. For example, the Fed chair can preapprove a general schedule of swap drawings and delegate individual drawing approvals consistent with that schedule to the FRBNY (FOMC 2021a). Under the Fed swap line to Mexico in 1994, all draws were subject to approval by the FOMC. However, the Committee "activated" part of the line when approved, meaning that the Mexican central bank could immediately draw on it up to that limit without further FOMC approval (Fed 1995, 12). The Fed used this tactic again during the GFC. In December 2007, at the same time the FOMC approved a USD 20 billion swap line for the ECB, it preauthorized the ECB to draw up to USD 10 billion under the line. Draws in excess of that amount required additional FOMC approval (FOMC 2007b). In October 2008, the FOMC also preapproved drawings up to USD 5 billion for the four emerging market central banks (BCRA, Bank of Korea [BOK], Banxico, MAS), allowing them to draw this much without additional FOMC approval (FOMC 2008).

For the CMIM, a committee of the deputies of the ASEAN + 3 central banks and finance ministries is responsible for approving draws under constraints (size limits, duration, renewals) set out in the agreement signed by the governors and finance ministers (AMRO n.d.).

Repos follow traditional processing procedures that facilitate efficient funding.

The ECB has not disclosed many details about the agreements entered into under the EUREP facility or how draws under this facility operate. However, for its bilateral repo arrangements with MNB and NBP, it uses a common agreement for financial transactions published by the Banking Federation of the European Union (EBF 2004), a Master Agreement for Financial Transactions, which it customizes (Gupta 2023a; Gupta 2023b). Use of a common form agreement indicates the parties' intent to adopt common transactional practices that facilitate efficient processing. The agreement also indicates that that Deutsche Bundesbank, the German central bank, would be responsible for securities settlement and collateral management for the ECB (ECB and MNB 2008). In general, the Bundesbank received a request for a draw and processed both sides of the transaction by valuing the collateral, confirming transfer of the cash, and allocating to the ECB's and Borrower's respective accounts.

A borrowing central bank approved under the Fed's FIMA Repo Facility can draw on the facility by sending a trade request to the FRBNY, which sends back a confirmation. Then, the Borrower sells the required Treasury collateral to the Fed in a repo transaction and the FRBNY moves the collateral from the Borrower's custody account to the Fed's portfolio account (Kelly 2023). After the collateral is received, the Fed sends the haircut-adjusted cash value of the securities to the Borrower (Choi et al. 2022). The FRBNY, rather than the triparty repo system (provided by Bank of New York Mellon in the US) or an outside vendor, is responsible for post-trade clearing, settlement, and collateral management (Choi et al.

2022). Using the Fed's own systems rather a private market participant's may be beneficial for maintaining confidentiality of Borrower activity (Potter 2017).

10. Downstream Use of Borrowed Funds: The predominate use of borrowed funds is for downstream liquidity provision to banks in the borrowing central bank's jurisdiction. The lending central bank often informally influences the downstream use of funds by the borrowing central bank, and coordination between the lending and borrowing central banks is common.

In most cases, the borrowing central bank uses the funds to provide downstream liquidity in the foreign currency to banks in its jurisdiction at terms similar to the swap line. This has become one of the most predominate uses of swaps during the last two decades (McCauley and Schenk 2020) (see Key Design Decision No. 1, Purpose and Type).

Under the swap or repo agreement, the borrowing central bank typically has sole discretion over the use of currency borrowed through a swap. That was the case in 2010 when the BoE entered into a swap arrangement with the ECB to provide GBP 10 billion. That agreement did not limit how the funds could be used, but the ECB announced that the funds would be used only to provide liquidity to the CBIR to lend to its domestic banks (Wiggins 2023).

In an atypical situation, the medium-term swaps under the US–Mexico 1995 assistance package differed from the US's previous assistance, which had been only short-term bridge funding to facilitate Mexico's securing an IMF package. The 1995 package was intended to facilitate a more permanent solution for Mexico's problems and specified that the proceeds were to be used to "restructure Mexico's debt to longer terms" (Swaminathan and Wiggins 2023). To oversee this requirement, in addition to the FRBNY's approving the typical transactional terms, any requested draws also had to be approved by the FOMC, which could query the Banxico's intended use (Swaminathan and Wiggins 2023).

Specific restrictions on downstream use of drawn funds are rare. However, as a practical matter, the lending central bank can exercise some informal influence over intended uses by the Borrower because of its ability to deny requested draws, and the timing and terms of downstream facilities generally match the timing and terms of the swap transactions between the central banks.

Downstream facilities are typically structured as repos, providing liquidity in the Lender's currency secured by collateral denominated in the Borrower's currency. The Borrower determines eligibility, acceptable collateral, any haircut, and fees. It would typically demand the same high-quality collateral that it would demand in its other emergency liquidity facilities and apply similar haircuts (Bahaj and Reis 2018). There is no contractual relationship between the lending central bank and the downstream banks in the Borrower's jurisdiction. From the lending central bank's point of view, it has a mechanism to extend the reach of its currency to companies in the foreign jurisdictions without assuming the administrative duties or the risk of the foreign commercial banks.

The timing of the downstream repos generally matches the maturity of the swap draws. In our Eurozone–UK 2019 case, the connection between the BoE's draws of euros under its standing swap line with the ECB and its weekly downstream lending of euros under its new Liquidity Facility in Euros (LiFE) program were obvious—the LiFE program was "facilitated by the activation of the standing swap line" (BoE 2019). This operation mirrored an existing weekly US dollar facility that relied on a similar swap arrangement between the BoE and the Federal Reserve (Swaminathan 2023a).

The parties often coordinate or cooperate on the use of borrowed funds. The Fed, in particular, has a history of coordinating the deployment of funds borrowed under its dollar-liquidity swap lines. In 1967, the Fed cooperated with the BIS, Deutsche Bundesbank, and other central banks to provide US dollar liquidity to European banks to address weaknesses in the Eurodollar market related to the drop in the pound sterling (Arnold 2023e).

In December 2007, the Fed set up its earliest GFC swap lines with the SNB and ECB, and the parties coordinated to provide US dollar liquidity overseas at terms similar to those of the Fed's domestic Term Auction Facility (Runkel 2022). Fed officials believed using the complementary swap lines would help guard against spillover effects in American markets and reduce US dollar term funding pressures for European banks (French 2023b). The borrowing central banks worked closely with the Fed to structure downstream lending facilities for financial institutions in their jurisdictions (Fleming and Klagge 2010).

In the COVID-19 pandemic, the Fed enhanced its standing swap lines and established new temporary US dollar-liquidity swap lines. The borrowing central banks again used the proceeds of swap drawings to lend US dollars downstream to their domestic banks, conducting operations following a schedule preapproved by the Fed (Choi et al. 2022). The borrowing central banks determined the other terms of the downstream lending operations, including the method of allocation, eligible institutions, and collateral requirements (Bahaj and Reis 2020; Fed 2020c). Coordinating with borrower banks in this manner gave the Fed more control over the flow of US dollar liquidity and helped prevent arbitrage.

Most cases exhibited a similar connection between the swap line and the downstream lending facility (for example, Eurozone–Ireland and UK 2010 and Switzerland–Multiple 2008–09). The announcement of the swap or repo arrangement adds credibility to the borrowing central bank's downstream lending facility by making it clear that it has an available supply of the needed foreign currency, as the MNB expressed in its discussions with the ECB prior to finalizing its repo arrangement; it expected a "positive signaling effect" from announcing the ECB arrangement (ECB 2008a, 2).

During the GFC, the ECB, MNB, and NBP offered Swiss francs to banks in their jurisdictions via weekly auctions that coincided with the SNB's own swap auctions for banks that were its repo counterparties (Allen and Moessner 2010; SNB 2010b).

11. **Duration of Swap Draws and Repos: The duration of swap draws and repos is usually set out in the governing agreement or framework and in our cases varied**

from overnight to 95 days. The parties agree on the actual duration of each draw or repo at the time it is made. Draws also may be rolled over for longer periods.

Swap and repo agreements usually address the duration of drawings by (i) setting a specific maturity for swap draws (for example, "30 days"), (ii) setting a maximum term for the draw (for example, "of not more than X days"), or (iii) providing that the parties will agree on the duration at the time of the draw. Most swap and repo agreements utilize one of these mechanisms, allowing the parties to agree on the actual terms at the time of the draw. We also see central banks setting a duration administratively before actual draws are made.

The initial periods of actual draws ranged from overnight to 95 days (this excludes swaps with longer maturities found in Asia and Latin America; see Figure 10). However, it is useful to note that some swap and repo lines do allow draws to be rolled over, extending the actual length of the draw.

In some cases, the two central banks tailored the durations of their swap draws to fit the needs of the situation at hand. For instance, the agreements of the Fed's standing dollar swap lines with major central banks specify that the parties will agree on the swap duration when negotiating a draw (FRBNY n.d.c). Typically, the Fed provides swap draws of seven-day maturities to match the maturities of corresponding downstream dollar auctions. During crises, like the SDC or the COVID-19 pandemic, the Fed's swap lines provided 84-day maturities as major central banks announced the addition of 84-day maturity downstream US dollar operations (Fed 2020a; FRBNY 2021).

Liquidity constraints are often characterized by shortened maturities. In a prolonged crisis such as the GFC, lengths of draws may be extended in response to such market conditions. During the GFC, draws by various central bank counterparties under the Fed's swaps ranged from one to 95 days (French 2023b). Initially, the Fed and counterparties coordinated the downstream lending made possible by the swaps with the Fed's domestic lending under the Term Auction Facility (French 2023b; Runkel 2022). Original terms were 28 days, and then the Fed authorized 84-day draws; several foreign central banks also began to extend longer-term US dollar loans in their downstream lending (French 2023b; Runkel 2022). The borrowing central banks first lent the drawn funds in one-month and three-month fixed-rate auctions at the lowest rate at which bids were accepted in the most recent TAF auctions. Starting in October 2008, when the Fed removed the size limits on its swaps with major European central banks and the BOJ, those central banks primarily lent drawn funds to banks in fixed-rate tenders with full allotment at one-week, one-month, and three-month maturities, at rates no longer tied to the rates under the TAF (Fleming and Klagge 2010).

The SAARC framework provides swap draws with an initial maturity of three months. Each swap draw can be rolled over twice (RBI 2012). Under the framework, the interest rate charged was the London Interbank Offered Rate (LIBOR) plus 200 bps for US dollar and euro swaps, and the RBI repo rate minus 200 bps for Indian rupee swaps (RBI 2012). At the second rollover of the swap, the interest rate would increase by 50 bps over the previously charged interest rate (Gupta 2023e; RBI 2012).

Figure 10: Duration of Swap Draws and Repos

Case	Swap/Repo Arrangement	Permitted Duration	Draw Renewal Information
ASA 1977	ASA 1977	1, 2, 3, or 6 months	Renewals permitted for maturities less than 6 months
ASEAN +3 CMIM 2010	ASEAN +3 CMIM 2010	ILP: 1 year; IDLP: 6 months[A]	ILP: flexible; IDLP: 3 renewals[A]
China–Argentina 2014	China–Argentina 2014	Varied[B]	No information
China–Hong Kong 2009	China–Hong Kong 2009	No information	No information
China–Mongolia 2011	China–Mongolia 2011	3–6 months; in 2014, extended up to 12 months	Renewed annually[C]
Eurozone–Denmark 2008	Eurozone–Denmark 2008	Up to 6 months	No information
Eurozone–EUREP 2020	Eurozone–EUREP 2020	No information	No information
Eurozone–Hungary Repo 2008	Eurozone–Hungary 2008	Up to 1 month	No information
Eurozone–Ireland and UK 2010	Eurozone–Ireland 2010	No information	No information
	Eurozone–UK 2010	No information	No information
Eurozone–Poland Repo 2008	Eurozone–Poland 2008	Up to 1 month	No information
Eurozone–Sweden 2007	Eurozone–Sweden 2007	Up to 3 months	No information
Eurozone–UK 2019	Eurozone–UK 2013	1 week	No information
SAARC 2012	India–Bhutan 2013	3 months	2 rollovers permitted
	India–Sri Lanka 2015	3 months	2 rollovers permitted
	India–Maldives 2019	3 months	2 rollovers permitted
Scandinavia–Iceland 2008	Denmark–Iceland 2008	1 month	Contracts renewed monthly
	Norway–Iceland 2008	1 month	Contracts renewed monthly
	Sweden–Iceland 2008	1 month	Contracts renewed monthly
Switzerland–Multiple 2008–09	Switzerland–Eurozone 2008	Determined at the time of the draw	Renewable on 2 days' notice
	Switzerland–Poland 2008	Determined at the time of the draw	No information
	Switzerland–Hungary 2009	Determined at the time of the draw	No information
US–BIS and Bundesbank 1967	US–Bundesbank 1962	No information	No information
	US–BIS 1965	1–3 months	Renewable up to 3 times
US FIMA Repo 2020	US FIMA Repo 2020	Overnight	Rolled over as needed
US–Mexico 1982	US–Mexico 1967	Overnight or 3 months	No information
	US–Mexico 1982 (Fed/UST)	Overnight or 3 months	Renewable 3 times, up to 1 year maximum
	US–Mexico 1982 (UST)	9 days	No information
US–Mexico 1994	US–Mexico 1994 (NAFA)[D]	3–11 months	Renewed twice
	US–Mexico 1994 (TSL)[D]	3 months	No information
	US–Mexico 1995 (AP)[D]	Medium-term swaps: 10–22 months	No information
US–Multiple 2001	US–Canada 2001	No information	No information
	US–Eurozone 2001	No information	No information
	US–UK 2001	No information	No information

Case	Swap/Repo Arrangement	Permitted Duration	Draw Renewal Information
US–Multiple 2007–09[E]	US–Eurozone 2007	Actual draws were 1 to 91 days	No information
	US–Switzerland 2007	No information	No information
	US–Australia 2008	Actual draws were 21 to 95 days	No information
	US–Brazil 2008	No information	No information
	US–Canada 2008	No information	No information
	US–Denmark 2008	Actual draws were 14 to 84 days	No information
	US–Japan 2008	Actual draws were 28 to 88 days	No information
	US–Korea 2008	Actual draws were 84 to 86 days	No information
	US–Mexico 2008	Actual draws were 88 days	No information
	US–New Zealand 2008	No information	No information
	US–Norway 2008	Actual draws were 28 to 84 days	No information
	US–Singapore 2008	No information	No information
	US–Sweden 2008	Actual draws were 84 to 88 days	No information
	US–UK 2008	Actual draws were 1 to 84 days	No information
US–Multiple 2010–11	US–Canada 2010	Up to 88 days	No information
	US–Eurozone 2010	Up to 88 days	No information
	US–Japan 2010	Up to 88 days	No information
	US–Switzerland 2010	Up to 88 days	No information
	US–UK 2010	Up to 88 days	No information
US–Multiple 2020	US–Canada 2013	Up to 88 days	No information
	US–Eurozone 2013	Up to 88 days	No information
	US–Japan 2013	Up to 88 days	No information
	US–Switzerland 2013	Up to 88 days	No information
	US–UK 2013	Up to 88 days	No information
	US–Australia 2020	Up to 88 days	No information
	US–Brazil 2020	Up to 88 days	No information
	US–Denmark 2020	Up to 88 days	No information
	US–Korea 2020	Up to 88 days	No information
	US–Mexico 2020	Up to 88 days	No information
	US–New Zealand 2020	Up to 88 days	No information
	US–Norway 2020	Up to 88 days	No information
	US–Singapore 2020	Up to 88 days	No information
	US–Sweden 2020	Up to 88 days	No information

Note: Fed/UST: Federal Reserve/US Treasury; UST: US Treasury: NAFA: North American Framework Agreement; TSL: Temporary Swap Line; AP: assistance package of 1995.

[A] The CMIM has two lending programs, the CMIM-SF, its standard crisis facility, and the CMIM-PL, a precautionary line. The swap terms differ for borrowers with and without a corresponding IMF program in place; swaps, therefore, are divided into an IMF-linked portion (ILP), and an IMF-delinked portion (IDLP). As opposed to the CMIM-SF, which can be drawn upon directly, the CMIM-PL provides borrowers with a temporary swap line that expires after six months. The duration of draws varies for each program. Swap duration for CMIM-SF, ILP: one year and flexible renewals (no defined limit); IDLP: six months and up to three renewals. Swap duration for CMIM-PL, ILP: one year and no renewals; IDLP: six months and no renewals. Line duration for CMIM-PL, ILP: six months and flexible renewals: IDLP: six months and three renewals.

[B] We uncovered limited information about the duration of swap transactions between the Central Bank of Argentina and the PBOC: the 2017 swap line had a repayment period of one year; the 2018 supplementary agreement had a repayment period of up to 370 days; and in 2020, at least one draw was made with a duration of three months.

[C] Based on the Bank of Mongolia's reports, it may have rolled its debt under the swap line each year, by paying down the line and then drawing the same amount, a common pattern.

[D] US aid to Mexico during its 1994–95 crisis involved several swap lines—(1) a temporary swap arrangement established by the Federal Reserve and the Treasury with the Bank of Mexico in March 1994, which the Banxico did not draw upon (ST Swap); (2) a trilateral network swap arrangement under the North American Framework Agreement among the US, Mexico, and Canada, which the Banxico did draw upon and which remains in place (NAFA); and (3) in February 1995, a USD 20 billion assistance package, which included the existing short-term ST Swap and NAFA swap lines, medium-term swaps (funded by the UST), and guarantees (AP).

[E] Unlike the Fed's swap arrangements during the Sovereign Debt Crisis and COVID-19 pandemic, we have not identified stated maximum durations in the Fed's GFC swap lines. Instead, we have provided a range for the actual durations for drawings under each swap line.

Sources: Authors' analysis; Arnold 2023b (b); Arnold 2023c (c); Fed n.d.b. (e); Hoffner 2023b (a); Horn et al. 2023 (c); Swaminathan and Wiggins 2023 (d).

12. **Rates and Fees: The parties agree on the exchange rates, determined by prevailing spot rates, at the time of the swap draw. There is no direct exchange rate risk because the spot and forward rates under a swap draw are the same, excluding interest.**

The parties generally agree on the exchange rate when one party wishes to draw on the swap. All the swap lines we examine in the survey adhere to a similar exchange rate convention—the exchange of two currencies at a market spot rate followed by an unwinding of the swap at the same initial exchange rate. The lending central bank is exposed to a credit risk, in the event the borrowing central bank is unable to repay. This risk is mitigated by having the swap be fully collateralized at the time of the initial swap draw. In this sense, the exchange rate risk is embedded in the credit risk. A devaluation of the collateral currency with respect to the lending currency could make repayment of the swap difficult.

Interest rates under the swaps are generally based on prevailing market rates and are usually the result of a formula and spread embodied in the swap agreement.

The interest rate is generally mutually agreed to by the parties as a transaction detail when a party wishes to draw under the swap. In a typical swap arrangement, the lending central bank charges the borrowing central bank an interest rate on drawings, with no interest charged on the collateral currency deposited by the Borrower (Bahaj and Reis 2021).

In our cases, we find less information with respect to interest rate calculations compared to other swap terms (such as size). In almost all surveyed swap lines that disclosed the interest-rate calculation, the formula always used a benchmark market reference rate (LIBOR, overnight index swap [OIS] rate) plus a premium. In these swap lines, the reference rate is an unsecured, interbank market rate. The CMIM is an exception, which uses the Secured Overnight Funding Rate (SOFR), as the reference rate. See Figure 11.

In our US–Mexico 1994 case, the US provided Mexico with short-term swaps (three months or less) and medium-term swaps that could carry maturities up to five years. The short-term swap facilities charged a 91-day Treasury bill (T-bill) rate, while the medium-term swap charged a 91-day T-bill rate plus a risk premium of 225–375 bps (Clinton 1995; US General Accounting Office 1996). The premium for medium-term swaps was set as the greater of "(1) a rate determined by the U.S. government's Interagency Country Risk Assessment System (ICRAS) to be adequate compensation for sovereign risk of countries such as Mexico or (2) a rate based on the amount of U.S. funds outstanding to Mexico from short- and medium-term swaps and securities guarantees at the time of disbursement" (US General Accounting Office 1996, 18).

By pricing swap lines at a premium, central banks ensure that usage largely remains limited to a crisis and that these facilities do not crowd out private-market instruments under normal circumstances (Goldberg and Ravazzolo 2021a; Singh 2023). These premiums ranged anywhere from 25 bps (OIS plus 25 bps in the case of Fed's COVID-19 swaps and LIBOR plus 25 bps in the ASA) to 200–400 bps. In our cases, these larger premiums show up in swaps to emerging market countries: China–Mongolia 2011 (Shanghai Interbank Offered

Rate [SHIBOR] + 200 bps), the SAARC swaps (three-month LIBOR + 200 bps), and for the US dollar swap leg of the China–Argentina 2014 swaps (SHIBOR + 400 bps).

In the China–Argentina 2014 swap line, the PBOC charged the BCRA 600–700 bps in interest (no formula given[24]) for renminbi draws secured by Argentinean pesos. The parties also established a supplementary agreement that allowed the BCRA to convert a portion of the renminbi drawn under the PBOC swap line into US dollars directly through the PBOC to boost its US dollar reserves to help it manage its exchange rate. The supplementary agreement to the swap line stated that the interest rate applicable to the BCRA's swaps of renminbi for US dollars was SHIBOR + 400 bps (BCRA 2015a; BCRA 2015b).

The ECB said that EUREP, its broad-based repo facility, was priced "slightly more expensive" than pricing under its bilateral repo arrangements (such as with the MNB and NBP) or swap lines. (ECB 2022b). The ECB's goal was to make the facility's pricing attractive only under adverse market conditions (ECB 2022b). ECB bilateral repo lines generally featured a higher lending rate than its swap lines (Albrizio et al. 2022). ECB repo lines (including the bilateral repo lines) generally applied haircuts to collateral (Panetta and Schnabel 2020).

In the Fed's US dollar–liquidity swap arrangements, we observe a connection between the interest rates under the swap draws and the rates occurring in the borrowing central bank's downstream lending operations. We do not observe this connection among other swap arrangements.

The only cases for which we find a direct connection between the swap interest rate and the rates charged for downstream lending are the Fed cases. Toward the beginning of the GFC, when the Fed worked with other major central banks in establishing new US dollar swap lines, the ECB wanted to structure its swap line so that it was perceived as a passive agent of the Fed in providing US dollar liquidity to Europe; it was important for the ECB to convey to the market that the underlying financial vulnerabilities were endemic to the US (Bernanke 2015, 183–84; Sheets 2019). This preference of the ECB's showed up in the relationship between the pricing of the draws on the Fed's swap lines and the ECB's downstream US dollar–liquidity operations. The ECB would lend US dollars to European financial institutions and remit all proceeds to the Fed, even if the interest from downstream lending exceeded the interest rate under the Fed swap. (FRBNY n.d.b; Sheets 2019). The Fed welcomed this pricing arrangement, as it allowed the Fed to earn more than it might have had the ECB just remitted the interest due under the swap agreement (Sheets 2019). This linked pricing scheme also appeared in the Fed's COVID-19 swaps lines. During the COVID-19 crisis, the four major central banks with standing swap lines at the Fed that conducted downstream US dollar operations (BoE, BOJ, ECB, and SNB) all lent out dollar liquidity in their jurisdictions at a fixed rate, full allotment, with the fixed rate set at the Fed swap line interest rate (OIS + 25 bps).

[24] We have not identified the interest formula for the renminbi draws on the 2014 PBOC–BCRA swap line. However, the 2009 PBOC–BCRA swap line had an interest rate of SHIBOR + 600 bps, for reference (Urien 2022).

Unlike these uncapped standing Fed swap lines, the Fed's temporary swap lines with nine central banks were capped at USD 30 billion to USD 60 billion, depending on the central bank. As a result, the central banks with temporary swap lines conducted downstream US dollar lending at competitive bid auctions, setting the minimum bid rate at the Fed swap interest rate (OIS + 25 bps). Following the dollar auction, the foreign central banks remitted the interest proceeds to the Fed (Bahaj and Reis 2018; 2020; FRBNY n.d.a). The difference in interest paid to the Fed under the standing swap lines and under the temporary swap lines, therefore, reflects the pricing differentials in downstream lending.

Managing interest rates can optimize the effectiveness of swap lines and downstream lending.

Lowering interest rate premiums on existing swap lines can be a quick tool to enhance liquidity provision to foreign funding markets. This is especially the case in Fed swaps where swap rates are directly tied to downstream lending. For instance, on March 15, 2020, at the onset of the COVID-19 pandemic, the Fed and five major central banks took coordinated action to improve USD liquidity in their jurisdictions. The Fed lowered the interest rate on its standing swap lines with the five banks (from OIS + 50 bps to OIS + 25 bps).[25] The lowered swap rate enabled these central banks to lend dollars in their jurisdictions at the lower rate. Following the first downstream US dollar operations at the reduced rate, dollar funding costs eased in those jurisdictions (Bahaj and Reis 2020).

High premiums on swap lines may produce stigma (particularly in connection to downstream lending operations). In November 2011, during the Sovereign Debt Crisis, Fed swap lines saw minimal use, attributed to significant stigma issues (FOMC 2011a). Such stigma undermines the effectiveness of swap lines to backstop US dollar funding abroad. In order to destigmatize the facility, the FOMC decided to reduce the premium from OIS + 100 bps to OIS + 50 bps (Bahaj and Reis 2022b; FOMC 2011a). Immediately after the reduction, usage increased with greater demand at downstream US dollar operations with better pricing encouraging broader participation in these offshore dollar operations (in Europe and Japan) (FOMC 2011b). The announcement of the reduction on November 30, 2011, also corresponded with a reduction in deviations from the covered interest rate parity; such deviations are indicators of rising premiums for dollar funding through foreign exchange swap markets (Bahaj and Reis 2022b; FOMC 2011b).

[25] *The FOMC manages its swap lines according to the procedures outlined by the Authorization for Foreign Currency Operations, which allows the chair of the FOMC to change the terms (for example, rates) of the swap lines without requiring FOMC approval* (FOMC 2020a; FOMC 2020d)*. The establishment of new swap lines, however, requires FOMC approval* (FOMC 2020a)*.*

Figure 11: Rates and Fees under Swap and Repo Arrangements

Case	Swap/Repo Arrangement	Interest Rate Calculation	Interest Rate Paid	Fees
ASA 1977	ASA 1977[A]	LIBOR + 25 bps[A]	No information	+100 bps fee for unpaid balance
ASEAN +3 CMIM 2010	ASEAN +3 CMIM 2010	SOFR + 150 bps[B]	No usage	+50–150 bps for renewals[B]
China–Argentina 2014	China–Argentina 2014	No information	600–700 bps; SHIBOR + 400 bps for RMB–USD secondary swap	None
China–Hong Kong 2009	China–Hong Kong 2009	No information	No information	None
China–Mongolia 2011	China–Mongolia 2011	SHIBOR + 200 bps	No information	None
Eurozone–Denmark 2008	Eurozone–Denmark 2008	ECB euro swaps rate + 75 bps	No information	No information
Eurozone–EUREP 2020	Eurozone–EUREP 2020	No information	No information	None
Eurozone–Hungary Repo 2008	Eurozone–Hungary 2008	1% initial margin ratio for intraday/overnight; 2% for longer	Redacted	No information
Eurozone–Ireland and UK 2010	Eurozone–Ireland 2010	No information	No information	No information
	Eurozone–UK 2010	No information	No information	No information
Eurozone–Poland Repo 2008	Eurozone–Poland 2008	1% initial margin ratio for intraday/overnight; 2% for longer	Redacted	No information
Eurozone–Sweden 2007	Eurozone–Sweden 2007	ECB euro swaps rate	Redacted	No information
Eurozone–UK 2019	Eurozone–UK 2013	No information	No information	No information
SAARC 2012	India–Bhutan 2013	USD, EUR: 3-mo. LIBOR + 200 bps; INR: RBI repo rate – 200 bps	No information	+50 bps interest for any second rollover
	India–Sri Lanka 2015	USD, EUR: 3-mo. LIBOR + 200 bps; INR: RBI repo rate – 200 bps	2.32% (2015)	+50 bps interest for any second rollover
	India–Maldives 2019	USD, EUR: 3-mo. LIBOR + 200 bps; INR: RBI repo rate – 200 bps	2.5%	+50 bps interest for any second rollover
Scandinavia–Iceland 2008	Denmark–Iceland 2008	No information	Roughly 3%	No information
	Norway–Iceland 2008	No information	Roughly 3%	No information
	Sweden–Iceland 2008	No information	Roughly 3%	No information
Switzerland–Multiple 2008–09	Switzerland–Eurozone 2008	Agreed at time of transaction[C]	No information	None
	Switzerland–Poland 2008	Agreed at time of transaction[C]	No information	None
	Switzerland–Hungary 2009	Agreed at time of transaction[C]	No information	None
US–BIS and Bundesbank 1967	US–Bundesbank 1962	No information	No information	None
	US–BIS 1965	BIS: Eurodollar rate – 100 bps	Approx. 4.75% to 5.75%	None
US FIMA Repo 2020	US FIMA Repo 2020	Interest rate on excess reserves + 25 bps	35 bps	None
US–Mexico 1982	US–Mexico 1967	No information	T-bill rate	None
	US–Mexico 1982 (Fed/UST)	No information	T-bill rate for Fed swaps; no information for UST	None
	US–Mexico 1982 (UST)	No information	No information	USD 50 million negotiating fee
US–Mexico 1994	US–Mexico 1994 (NAFA)	No information	91-day T-bill rate	No information
	US–Mexico 1994 (TSL)	No information	91-day T-bill rate	No information
	US–Mexico 1995 (AP)	Medium-term swaps: 91-day US T-bill rate + 225–375 bps	91-day US Treasury-bill rate + a risk premium of 225–375 bps	No information
US–Multiple 2001	US–Canada 2001	No information	No information	None
	US–Eurozone 2001	Agreed at time of transaction	No information	None
	US–UK 2001	No information	No information	None

Case	Swap/Repo Arrangement	Interest Rate Calculation	Interest Rate Paid	Fees
US–Multiple 2007–09	US–Eurozone 2007	Based on market rate	0.25% to 11.96%	None
	US–Switzerland 2007	Based on market rate	0.14% to 5.10%	None
	US–Australia 2008	Based on market rate	0.76% to 3.49%	None
	US–Brazil 2008	Based on market rate	No usage	None
	US–Canada 2008	Based on market rate	No usage	None
	US–Denmark 2008	Based on market rate	0.70% to 2.35%	None
	US–Japan 2008	Based on market rate	1.15% to 4.14%	None
	US–Korea 2008	Based on market rate	0.76% to 6.84%	None
	US–Mexico 2008	Based on market rate	0.70% to 0.74%	None
	US–New Zealand 2008	Based on market rate	No usage	None
	US–Norway 2008	Based on market rate	0.79% to 3.60%	None
	US–Singapore 2008	Based on market rate	No usage	None
	US–Sweden 2008	Based on market rate	0.14% to 5.10%	None
	US–UK 2008	Based on market rate	1.00% to 5.21%	None
US–Multiple 2010–11	US–Canada 2010	OIS + 100 bps until Nov. 2011; OIS + 50 bps after	No usage	None
	US–Eurozone 2010	OIS + 100 bps until Nov. 2011; OIS + 50 bps after	0.57% to 1.25%	None
	US–Japan 2010	OIS + 100 bps until Nov. 2011; OIS + 50 bps after	0.58% to 1.25%	None
	US–Switzerland 2010	OIS + 100 bps until Nov. 2011; OIS + 50 bps after	0.57% to 1.10%	None
	US–UK 2010	OIS + 100 bps until Nov. 2011; OIS + 50 bps after	No usage	None
US–Multiple 2020	US–Canada 2013	OIS + 25 bps	No usage	None
	US–Eurozone 2013	OIS + 25 bps	0.29% to 0.45%	None
	US–Japan 2013	OIS + 25 bps	0.29% to 0.41%	None
	US–Switzerland 2013	OIS + 25 bps	0.29% to 0.45%	None
	US–UK 2013	OIS + 25 bps	0.29% to 0.45%	None
	US–Australia 2020	OIS + 25 bps minimum[D]	0.30% to 0.34%	None
	US–Brazil 2020	OIS + 25 bps minimum[D]	No usage	None
	US–Denmark 2020	OIS + 25 bps minimum[D]	0.29% to 0.45%	None
	US–Korea 2020	OIS + 25 bps minimum[D]	0.29% to 0.91%	None
	US–Mexico 2020	OIS + 25 bps minimum[D]	0.32% to 0.91%	None
	US–New Zealand 2020	OIS + 25 bps minimum[D]	No usage	None
	US–Norway 2020	OIS + 25 bps minimum[D]	0.32% to 0.37%	None
	US–Singapore 2020	OIS + 25 bps minimum[D]	0.30% to 1.09%	None
	US–Sweden 2020	OIS + 25 bps minimum[D]	No usage	None

Note: Fed/UST: Federal Reserve/US Treasury; UST: US Treasury: NAFA: North American Framework Agreement; TSL: Temporary Swap Line; AP: assistance package of 1995.

[A] Information given refers to the 2005 memorandum of understanding.

[B] The CMIM uses the six-month term Secured Overnight Financing Rate plus a spread adjustment of 0.428% as the reference rate. The CMIM adds to the reference rate an initial premium of 150 bps, which increases by 50 bps for additional renewals up to a maximum premium of 300 bps.

[C] SNB swap transactions use the same exchange rate for the spot and forward rates but with a discount or premium to account for the spread between the interest rates on the two currencies, called the swap points. In effect, the borrowing central bank pays the rate of interest on the currency they borrow and receives the interest rate on the currency they lent in exchange. So, the spot exchange rate was the market spot rate, but the forward exchange rate was the spot exchange rate +/- forward swap points.

[D] Actual interest rates were based on downstream auction results. The Fed's counterparty central banks with temporary swap lines would provide dollars in their jurisdictions through competitive auctions and remit the proceeds to the Fed; as a result, the interest paid typically settled higher than the minimum OIS + 25 bps rate.

Sources: Authors' analysis; French 2023a (c); Hoffner 2023b (b); Hoffner 2023d (d).

13. Balance Sheet Protection: Under swap arrangements, Lenders are protected by holding the Borrower's currency as collateral; they do not transact directly with downstream foreign financial institutions. However, Lenders can take further steps to protect their balance sheets by including set-off provisions, customizing collateral requirements, or requiring that additional terms be met for a draw.

Central banks protect themselves with set-off rights, which can be plain vanilla or enhanced, permitting access to assets beyond the swap arrangement in the event of default.

Although the lending central bank faces limited credit risk because its counterparty is the borrowing central bank, it may still consider there to be some risk and try to mitigate it. One mechanism lending central banks use to protect against such an occurrence when providing some economic relief is to include a set-off clause in the swap or repo agreement.

A typical set-off clause gives the lender the right to seize assets of a defaulting party that it holds and deduct from such assets (set off) any amounts that are owed by the defaulting party, providing the Lender some economic relief and avoiding the situation where it has not been paid but would have to pay amounts or return assets to a defaulting party (Kagan 2020). In swap agreements between the Fed and the central banks of the Major Bank Swap Network, the scope of the set-off provisions is limited to defaults under the agreement and swap related assets under the agreement, and are mutual.[26] (See, for example, the Fed's agreements with the BOJ and the SNB [FRBNY 2011a; FRBNY 2011b].) For example, if a Borrower were to default on paying interest owed at the end of a swap draw, the Fed could deduct the amount of interest from the currency collateral before returning it.

Transcripts from FOMC meetings during the GFC provide additional insight into how Fed officials evaluated the risks of each counterparty central bank. In October 2008, the Fed discussed at length the approval of the last four central bank swap lines, to Brazil, Korea, Mexico, and Singapore. According to the transcripts, unlike the swap lines with major central banks, the Fed included enhanced, broad, set-off rights in these four swap arrangements. In the event of a default, the Fed would be able to not only seize the currency collateral but also take any assets of the defaulting central bank held at the FRBNY to satisfy amounts owed under the swap arrangement (FOMC 2008). As described by then–FRBNY President Timothy Geithner, these broad rights added valuable security to the arrangement for the Fed:

> These countries hold substantial amounts of their reserves in dollars. They hold a substantial fraction of those dollars in accounts at the New York Fed. If they defaulted on their piece of the swap and the falling value of their currency left us with some exposure, we would have the ability to take assets from their accounts to cover any loss. So, it's better than the fact that this is a sovereign credit and it is better than the fact that we have an asset on the other side of the swap, because they hold substantial foreign

[26] "In the event a party fails to fulfill its obligations under this Swap Agreement with respect to a particular Swap transaction on the applicable Maturity Date, the non-defaulting Party is authorized to set off any obligations they may owe the defaulting Party against the currency held by the non-defaulting Party pursuant to this Swap Agreement" (FRBNY 2011a; FRBNY 2011b, para. 5(a).

exchange reserves with us. (FOMC 2008, 20)

It is unclear whether or not these set-off rights, discussed in 2008, applied to the COVID-19-era swap lines. However, it is notable that although the Fed does disclose the complete agreements embodying the standing swap arrangements with major central banks, its policy is to not disclose the legal agreements for its temporary USD swap agreements (FRBNY n.d.b).

The US used a more robust set-off mechanism under the USD 20 billion 1995 assistance package to Mexico. The Mexican government was required to deposit revenues from oil exports and certain other amounts into an account at the FRBNY to which the Fed had certain set-off rights. Under the contemporaneous Oil Proceeds Facility Agreement, which was to remain in place until Mexico had satisfied all its obligations under any component of the 1995 package (short-term swaps, medium-term swaps, guarantees), if at any point Mexico defaulted on any of its obligations, the FRBNY could access such funds to satisfy the obligations (US Treasury 1995). See Swaminathan and Wiggins (2023) for more discussion of these and other protective elements.

Central banks may protect themselves by requiring a third-party currency as collateral from the borrowing central bank or by entering into a repo arrangement and accepting securities denominated in its currency, instead of the borrower's currency, as collateral.

Central banks usually choose to forego swap arrangements with central banks from countries whose currency they do not wish to accept as collateral. During the GFC, the Fed declined to enter into swaps with seven countries that requested them.[27] However, if the lending central bank does wish to enter into a swap agreement with a country whose currency it does not wish to accept, it has the option of requiring as collateral a currency other than that of the Borrower, usually an easily convertible currency from an advanced economy. The SNB did this under its GFC swap agreements with the NBP (2008) and the MNB (2009); it accepted only euros, not złoty or forints, in exchange for Swiss francs (French 2023a).

A central bank can also choose to enter into a repo agreement, which requires securities as collateral rather than currency. The ECB chose to enter into bilateral repo arrangements rather than the requested swap arrangements with the MNB and NBP, non–euro area countries. Under these agreements, the ECB accepted only certain euro-denominated government securities as collateral, the scope of which was separately negotiated with each party (Gupta 2023a; Gupta 2023b). In March 2020, the ECB again entered into several bilateral repo arrangements with non–euro area countries accepting only highly rated euro-denominated government securities (ECB 2022c).

The recently adopted EUREP and FIMA repo frameworks provide the ECB and Fed additional options in this regard. While the FIMA framework has standardized terms regarding collateral (accepting only US Treasuries), the EUREP framework allows the ECB to

[27] Sahasrabuddhe (2019) has identified seven central banks that requested swap lines during the GFC but which the FOMC declined: Chile, the Dominican Republic, India, Iceland, Indonesia, Peru, and Turkey.

individually negotiate terms with each counterparty, but the ECB has stated that the range of collateral is narrower than for a bilateral repo line (Arnold 2023d; Kelly 2023). (See Key Design Decision No. 1, Purpose and Type.)

Central banks can protect themselves by further securing the value of collateral currency with a top-up clause.

If there is some risk that the borrowing central bank's currency may be devalued over the life of the draw, the lending central bank can include a "top-up clause" that requires the Borrower to deposit additional collateral with the Lender if the value of its currency decreases against the borrowed currency. The United States included this type of provision in its medium-term swap line with the Mexico in 1994, which allowed for longer than usual maturities of up to five years. Mexico bore any exchange rate risk, protecting the US from losses on movements in the prevailing exchange rate (Clinton 1995). The parties calculated the applicable exchange rate on a quarterly basis and, in the event of an unexpected appreciation of US dollars against pesos, Mexico would have to adjust the peso collateral to reflect the prevailing exchange rate (Clinton 1995). The ECB included similar clauses in its GFC-era bilateral repo agreements with MNB and NBP (Gupta 2023a; Gupta 2023b). However, most swap agreements do not contain such a top-up clause.

Central banks protect themselves by having the fiscal authority guarantee swap transactions.

When the US extended swap assistance to Mexico in 1982 and 1994, the swaps were partially funded by the Fed and partially by the Treasury. The swap arrangements also permitted longer than usual maturities, up to three months under the joint Fed-Treasury swaps in 1982 and up to five years under the 1995 US Treasury swaps from the multilateral assistance package. In both instances, there were questions about whether Mexico had sufficient collateral to support the swap line. FOMC minutes also reveal that the Committee had some concern that it was being asked to perform quasi-fiscal actions (FOMC 1994). Therefore, in 1982, the Treasury guaranteed the Fed drawings of three months, and in 1995, it guaranteed drawings that were outstanding for more than 12 months (Swaminathan 2023b; Swaminathan and Wiggins 2023; US General Accounting Office 1996).

Central banks can protect themselves by requiring additional administrative requirements to draw under the swap arrangements or by applying additional limitations and scrutiny to each draw.

The lending central banks in our cases used a variety of other mechanisms to minimize risk under swap and repo arrangements and protect their balance sheets. These efforts included: limiting the amounts that could be borrowed without additional review; requiring that the Borrower maintain additional collateral at the Lender; shrinking the amount permitted to be borrowed under the arrangement as it was used; and requiring additional commitments from the Borrower's fiscal authority regarding use of funds or fiscal integrity, adherence to certain additional financial standards, and the production of a certification from a third party such as the IMF.

For example, as part of a request for a swap draw under the ASEAN network, the member must attach an assessment of its domestic economic situation and balance of payments outlook (ASEAN 2005; Hoffner 2023a). The Agent Bank (the ASA administrator) must confirm satisfaction of this requirement and coordinate processing the request among network members (Hoffner 2023a).

Although designed to further protect the Lender, there is no guarantee that the Borrower will adhere to such additional conditions. A Lender can protect itself in the event that the Borrower does not comply, however, by retaining the option to not approve a requested draw in such circumstance. For example, under the Scandinavia–Iceland swaps, the Icelandic government made certain commitments to the Lenders in support of the swaps, and the CBI was required to update them on the government's adherence to the commitments. The then–governor of the Riksbank, Stefan Ingves, who was present at the negotiations over the swap lines, likened these swap conditions to that of an IMF program, saying: "It was almost as if it was designed as an IMF program without the IMF." (Ingves 2022, 10). When the CBI sought to draw under the lines, the DNB and Norges Bank approved the draw, but the Riksbank did not because it felt the Icelandic government had not delivered on its commitments (Hoffner 2023c).

14. Other Restrictions: In a typical swap, there are no formal restrictions on the downstream use of borrowed funds.

Typically, the use of funds drawn under a swap is at the discretion of the borrowing central bank. However, in several cases, the lending central bank imposed informal restrictions on the downstream usage of borrowed funds through its ability to approve swap draws. See Key Design Decision No. 10, Downstream Use of Borrowed Funds.

15. Other Options: Lending central banks often prioritize swap arrangements over other alternatives owing to their ability to efficiently address international liquidity issues with minimal risk.

For lending central banks, swap arrangements are an efficient way to provide liquidity in their currencies to foreign jurisdictions with limited risk. While direct lending to foreign commercial banks could be an option, the Lenders would be exposed to foreign counterparty credit risks that are better managed and assessed by the borrowing central banks.

Repo arrangements, such as the ECB's bilateral repos and EUREP facility and the Fed's FIMA Repo Facility, are an increasingly popular alternative to swaps. These arrangements address the lending central bank's dilemma of providing liquidity abroad while being cautious about accepting the counterparty's currency as collateral due to potential sovereign default risks.

Borrowers with ample foreign currency reserves may be able to meet their own liquidity needs. Even if the borrowing central bank has substantial reserves, though, it may not wish to use them for its liquidity needs. This was a sentiment expressed by the MNB in its discussions with the ECB regarding the 2008 repo arrangement between the parties. The MNB specifically indicated that it would use the facility as a mechanism of "first resort" and would draw on the facility before using its euro reserves (ECB 2008a; MNB 2008). Still, as central banks typically hold reserves in government securities, even those countries may prefer to use an international repo facility to access liquidity without disrupting the global market for those securities.

Borrowers without sufficient reserves to cover their domestic banks' foreign currency liquidity needs may also have difficulty securing reserve currency swap arrangements and so look to the IMF's Short-term Liquidity Line and Flexible Credit Line (Hoffner 2023c; Negus 2020). (See Key Design Decision No. 7, Eligible Institutions.)

16. Lifespan of the Agreement and Exit Strategy: Swap agreements tend to have lifespans of 30 days to three years, with arrangements entered into for liquidity provision having shorter lifespans. Standing agreements, those without end dates, seem to occur only between advanced-economy countries.

Almost all of the swap and repo arrangements in our survey had explicit termination dates. Swap arrangements and repos created as liquidity-providing facilities tend to have shorter initial lifespans than swaps created for other purposes. However, during crises, central banks often extend the initial end date of swap arrangements, resulting in much longer lifespans.

The length of initial lifespans ranged from 30 days to three years. The most common initial lifespans were four to six months. Central banks don't provide much explanation about how they set the lifespans of swap arrangements. Based on the evidence in our survey, it appears that lending central banks typically sought to announce swaps with lifespans long enough that the public would perceive them to be credible and useful. In some cases, they announced shorter initial lifespans to signal that they thought a liquidity crisis wouldn't last long, such as the Fed's 30-day swap arrangements with the BoC, BoE, and ECB after the September 11 attacks (US–Multiple 2001).

In 10 instances, central banks indicated the initial lifespan of a facility with a qualifier such as, "at least until [date]") or "for at least six months" (see Switzerland–Eurozone 2008 and US–Multiple 2020, respectively). Arguably, these facilities also had definite expiration dates. Central banks apparently used this language to promise that there would be a further announcement about whether the facility would continue beyond the initial end date; in practice, they have usually kept that promise (see Key Design Decision No. 6, Communication).

Extension of initial end dates was a frequent and often repeated occurrence, particularly with arrangements that were enacted during protracted crises such as the GFC, SDC, and COVID-19, and as shown in Figure 12, a majority of the swap and repo arrangements were extended, some multiple times. During the GFC, the Fed originally announced that its swap arrangements with 14 central banks were temporary for periods of four to six months but later extended them several times as they became critical to the crisis response; the arrangements ended only in February 2010 as usage waned (US–Multiple 2007–09) (French 2023b).

To respond to the COVID-19 pandemic in March 2020, the Fed again enacted temporary swap lines with nine central banks to "be in place for at least six months" (Fed 2020b). The FOMC then extended the lines three times, saying that extension would "help sustain improvements in global U.S. dollar funding markets by serving as an important liquidity backstop" (Fed 2020h). The lines expired in December 2021, 21 months after origination.

The ECB implemented the EUREP in June 2020 with an initial lifespan of approximately one year and then extended it thrice, to January 2024, taking the view that the lines supported its monetary policy (ECB 2022a). The ECB also extended until January 2024 the swaps and repos that it implemented during the COVID-19 pandemic, in response to the war in Ukraine (ECB 2022c).

Rather than use a termination date, central banks in three cases announced that they would leave the arrangement in place "until countermanded" (Eurozone–Hungary Repo 2008, Eurozone–Poland Repo 2008) or "as long as needed" (Eurozone–Denmark 2008). Such language communicates that the facility is temporary but not strictly time bound. It frees the central bank from publicly acting to sustain the facility.

Swap arrangements entered for reasons other than liquidity provision tend to have definite end dates with longer initial lifespans. In the cases we reviewed, the lifespans of non-liquidity arrangements ranged from two to almost 30 years. These agreements were regularly reviewed by the parties and often extended, sometimes with adjustments made to the facility. For example, the US swap arrangement with Mexico originated in 1967 during the Bretton Woods era and was replaced only in 1994, when it was superseded by the NAFA (Swaminathan and Wiggins 2023).

A common practice of the PBOC (including the three arrangements that we analyzed) is to give its arrangements (entered into for trade purposes) lifespans of between two and three years and then subject each to a review before deciding to extend the agreement, often with modifications to the terms. For example, the PBOC and BCRA renewed their 2014 swap arrangement in 2017, expanded it significantly in 2018, and renewed it for an additional three years in 2020 (for a total lifespan of nine years), with the arrangement still continuing (Arnold 2023b). The PBOC's arrangement with the BOM, originated in 2011, was for three years but has been extended three times, for a total lifespan of 12 years, and was expanded in size (Arnold 2023c). Despite such long lifespans, the PBOC seems to prefer not to create standing arrangements.

Figure 12: Life Cycle of Swap or Repo Arrangement and Exit Strategy

Case	Swap/Repo Arrangement	Origination Date	Date of First Draw or Activation	Original End Date	Extended or Renewed, Other Action	Final End Date or Status
ASA 1977	ASA 1977	Aug. 5, 1977	1979	Aug. 5, 1978	Renewed multiple times	Nov. 16, 2021
ASEAN +3 CMIM 2010	ASEAN +3 CMIM 2010	March 24, 2010	No usage	Standing agreement	Standing agreement	Standing agreement
China–Argentina 2014	China–Argentina 2014	July 2014	Oct. 30, 2014	July 2017	Extended 2 times	Continuing
China–Hong Kong 2009	China–Hong Kong 2009	Jan. 20, 2009	Oct. 2, 2010	Jan. 20, 2012	Extended 5 times	Made standing in July 2022
China–Mongolia 2011	China–Mongolia 2011	May 6, 2011	2012	May 6, 2014	Extended 3 times	July 31, 2023
Eurozone–Denmark 2008	Eurozone–Denmark 2008	Oct. 27, 2008	2008–09	"As long as needed"	N/A	Standing agreement [A]
Eurozone–EUREP 2020	Eurozone–EUREP 2020	June 25, 2020	No information	June 2021	Extended 3 times	Jan. 15, 2024
Eurozone–Hungary Repo 2008	Eurozone–Hungary 2008	Oct. 16, 2008	Oct. 21, 2008	"Until countermanded"	Half of repo converted to a swap in 2010	Jan. 15, 2024
Eurozone–Ireland and UK 2010	Eurozone–Ireland 2010	Dec. 10, 2010	No information [B]	Sept. 30, 2011	Extended 1 time	Sept. 28, 2012
	Eurozone–UK 2010	Dec. 17, 2010	No information	Sept. 30, 2011	Extended 3 times	Sept. 30, 2014
Eurozone–Poland Repo 2008	Eurozone–Poland 2008	Nov. 6, 2008	No information	"Until countermanded"	Repo converted to "precautionary" swap in 2022	Jan. 15, 2024
Eurozone–Sweden 2007	Eurozone–Sweden 2007	Dec. 20, 2007	June 10, 2009	Standing agreement	N/A	Standing agreement
Eurozone–UK 2019	Eurozone–UK 2013	Oct. 31, 2013	March 13, 2019	Standing agreement	N/A	Standing agreement
SAARC 2012	India–Bhutan 2013	March 8, 2013	March 8, 2013	March 8, 2016	Extended multiple times	June 30, 2023
	India–Sri Lanka 2015	March 25, 2015	March 25, 2015	March 25, 2018	Extended multiple times	Standing agreement
	India–Maldives 2019	July 2019	April 2020	July 2020	Extended 2 times	Renewed on Dec. 8, 2022
Scandinavia–Iceland 2008	Denmark–Iceland 2008	May 16, 2008	Oct. 14, 2008	Dec. 31, 2008	Extended 1 time	Dec. 31, 2009
	Norway–Iceland 2008	May 16, 2008	Oct. 14, 2008	Dec. 31, 2008	Extended 1 time	Dec. 31, 2009
	Sweden–Iceland 2008	May 16, 2008	After Nov. 20, 2008	Dec. 31, 2008	Extended 1 time	Dec. 31, 2009
Switzerland–Multiple 2008–09	Switzerland–Eurozone 2008	Oct. 2008	Oct. 2008	"At least until Jan. 2009"	Extended 3 times	Jan. 2010
	Switzerland–Poland 2008	Nov. 2008	Nov. 2008	Jan. 2009	Extended 3 times	Jan. 2010
	Switzerland–Hungary 2009	Jan. 2009	Feb. 2009	April 2009	Extended 3 times	Jan. 2010
US–BIS and Bundesbank 1967	US–Bundesbank 1962	Aug. 1962	No information	No information	Renewed multiple times	Nov. 17, 1998
	US–BIS 1965	Aug. 1965	Nov. 13, 1967	No information	Renewed multiple times	Nov. 17, 1998
US FIMA Repo 2020	US FIMA Repo 2020	March 31, 2020	Week ended April 8, 2020	Sept. 30, 2020	Extended 2 times	Made standing on July 28, 2021
US–Mexico 1982	US–Mexico 1967	May 2, 1967	April 30, 1982	Standing agreement	N/A	Standing agreement
	US–Mexico 1982 (Fed/UST)	Aug. 28, 1982	Aug. 1982	Aug. 23, 1983	No	Aug. 23, 1983
	US–Mexico 1982 (UST)	Aug. 15, 1982	Aug. 15, 1982	No information	No	Repayment: Aug. 24, 1982; Expiration: Not stated

Case	Swap/Repo Arrangement	Origination Date	Date of First Draw or Activation	Original End Date	Extended or Renewed, Other Action	Final End Date or Status
US–Mexico 1994	US–Mexico 1994 (NAFA)	April 26, 1994	Jan. 11, 1995	Dec. 15, 1995	Renewed multiple times	Standing agreement
	US–Mexico 1994 (TSL)	March 24, 1994	No usage	April 29, 1994	No	April 29, 1994
	US–Mexico 1995 (AP)	Feb. 21, 1995	March 14, 1995	Jan. 16, 1997	No	Jan. 16, 1997
US–Multiple 2001	US–Canada 2001	Sept. 13, 2001	No usage	30 days after initiation	N/A	Expired 30 days after initiation
	US–Eurozone 2001	Sept. 12, 2001	Sept. 12, 2001	30 days after initiation	N/A	Expired 30 days after initiation
	US–UK 2001	Sept. 14, 2001	No usage	30 days after initiation	N/A	Expired 30 days after initiation
US–Multiple 2007–09	US–Eurozone 2007	Dec. 12, 2007	Dec. 21, 2007	June 12, 2008	Extended 5 times	Feb. 1, 2010
	US–Switzerland 2007	Dec. 12, 2007	Jan. 14, 2008	June 12, 2008	Extended 5 times	Feb. 1, 2010
	US–Australia 2008	Sept. 24, 2008	Sept. 26, 2008	Jan. 30, 2009	Extended 2 times	Feb. 1, 2010
	US–Brazil 2008	Oct. 29, 2008	No usage	April 30, 2009	Extended 1 time	Feb. 1, 2010
	US–Canada 2008	Sept. 18, 2008	No usage	Jan. 30, 2009	Extended 2 times	Feb. 1, 2010
	US–Denmark 2008	Sept. 24, 2008	Sept. 26, 2008	Jan. 30, 2009	Extended 2 times	Feb. 1, 2010
	US–Japan 2008	Sept. 18, 2008	Sept. 24, 2008	Jan. 30, 2009	Extended 3 times	Feb. 1, 2010
	US–Korea 2008	Oct. 29, 2008	Dec. 2, 2008	April 30, 2009	Extended 1 time	Feb. 1, 2010
	US–Mexico 2008	Oct. 29, 2008	April 21, 2009	April 30, 2009	Extended 1 time	Feb. 1, 2010
	US–New Zealand 2008	Oct. 28, 2008	No usage	April 30, 2009	Extended 1 time	Feb. 1, 2010
	US–Norway 2008	Sept. 24, 2008	Sept. 30, 2008	Jan. 30, 2009	Extended 2 times	Feb. 1, 2010
	US–Singapore 2008	Oct. 29, 2008	No usage	April 30, 2009	Extended 1 time	Feb. 1, 2010
	US–Sweden 2008	Sept. 24, 2008	Oct. 15, 2008	Jan. 30, 2009	Extended 2 times	Feb. 1, 2010
	US–UK 2008	Sept. 18, 2008	Sept. 18, 2008	Jan. 30, 2009	Extended 3 times	Feb. 1, 2010
US–Multiple 2010–11	US–Canada 2010	May 9, 2010	No usage	Jan. 31, 2011	Extended at least 3 times	Made standing in Oct. 2013
	US–Eurozone 2010	May 9, 2010	May 11, 2010	Jan. 31, 2011	Extended at least 3 times	Made standing in Oct. 2013
	US–Japan 2010	May 10, 2010	May 18, 2010	Jan. 31, 2011	Extended at least 3 times	Made standing in Oct. 2013
	US–Switzerland 2010	May 9, 2010	Aug. 10, 2011	Jan. 31, 2011	Extended at least 3 times	Made standing in Oct. 2013
	US–UK 2010	May 9, 2010	No usage	Jan. 31, 2011	Extended at least 3 times	Made standing in Oct. 2013

Case	Swap/Repo Arrangement	Origination Date	Date of First Draw or Activation	Original End Date	Extended or Renewed, Other Action	Final End Date or Status
US–Multiple 2020	US–Canada 2013	Oct. 31, 2013	No usage[C]	Standing agreement	Standing agreement	Standing agreement
	US–Eurozone 2013	Oct. 31, 2013	March 19, 2020[C]	Standing agreement	Standing agreement	Standing agreement
	US–Japan 2013	Oct. 31, 2013	March 19, 2020[C]	Standing agreement	Standing agreement	Standing agreement
	US–Switzerland 2013	Oct. 31, 2013	March 19, 2020[C]	Standing agreement	Standing agreement	Standing agreement
	US–UK 2013	Oct. 31, 2013	March 19, 2020[C]	Standing agreement	Standing agreement	Standing agreement
	US–Australia 2020	March 19, 2020	March 27, 2020	At least 6 months	Extended 3 times	Dec. 31, 2021
	US–Brazil 2020	March 19, 2020	No usage	At least 6 months	Extended 3 times	Dec. 31, 2021
	US–Denmark 2020	March 19, 2020	March 30, 2020	At least 6 months	Extended 3 times	Dec. 31, 2021
	US–Korea 2020	March 19, 2020	April 2, 2020	At least 6 months	Extended 3 times	Dec. 31, 2021
	US–Mexico 2020	March 19, 2020	April 3, 2020	At least 6 months	Extended 3 times	Dec. 31, 2021
	US–New Zealand 2020	March 19, 2020	No usage	At least 6 months	Extended 3 times	Dec. 31, 2021
	US–Norway 2020	March 19, 2020	March 30, 2020	At least 6 months	Extended 3 times	Dec. 31, 2021
	US–Singapore 2020	March 19, 2020	April 7, 2020	At least 6 months	Extended 3 times	Dec. 31, 2021
	US–Sweden 2020	March 19, 2020	No usage	At least 6 months	Extended 3 times	Dec. 31, 2021

Notes: Quoted text in this table comes from the original underlying agreements. Fed/UST: Federal Reserve/US Treasury; UST: US Treasury: NAFA: North American Framework Agreement; TSL: Temporary Swap Line; AP: assistance package of 1995.

[A] It is not clear if this arrangement was continuously in place from its origin in 2008. However, the ECB and DNB "reactivated" the line on March 20, 2022, and the ECB characterized it as a standing arrangement on its website.

[B] According to the Central Bank of Ireland's annual report, between the signing of the agreement in December 2010 and annual report's publication on May 20, 2011, one credit institution had used the downstream facility.

[c] On March 15, 2020, the Fed and the five central banks with standing US dollar swap lines announced enhancements to these lines that would go into effect for the swap operations during the week of March 16, 2020. The first drawings under the new enhancements settled on March 19, 2020.

Sources: Authors' analysis; CBIR 2011 (b); Fed n.d.b. (c); Fed 2020a (c).

Swap arrangements without stated termination dates were terminated through (a) deactivation, (b) pricing mechanisms, or (c) conversion to standing arrangements.

Using a fixed termination date for a swap arrangement is the most common exit strategy we observed among our swap lines and is particularly common for arrangements put in place to address perceived or occurring liquidity constraints. Deactivation of a standing swap line means that the use of the facility ceases but the facility continues to exist; it remains dormant until the next usage or activation. In deactivating a standing swap line, the parties simply announce that funding will be available until a certain date and then take no more action. For example, in 2021, the central banks of the Major Bank Swap Network wound down the frequency and tenure of swaps available under their lines.[28] (See Hoffner 2023d.)

In several cases, the parties used pricing as a natural exit mechanism. Lenders typically pegged swap rates to the Lender's policy rate, or a similar benchmark rate plus a margin, consistent with Bagehot's dictum urging central banks to lend freely but at a penalty rate (Bagehot 1873; Bahaj and Reis 2021; Tucker 2009). The penalty encourages banks to stop using swaps when a crisis ends, to avoid crowding out private market alternatives under normal market conditions (Choi et al. 2022). This exit mechanism is particularly relevant for standing swap lines such as the Major Bank Swap Network that have no set expiration date. For the Fed's standing swap lines, borrowing central banks may request swaps at any point. In practice, however, US dollar swap usage has remained almost entirely confined to periods of acute stress (for example, during the Sovereign Debt Crisis and COVID-19 pandemic) (Bahaj and Reis 2020). The ECB and Fed have both said that the rates charged under their framework repo facilities (EUREP and FIMA, respectively) are generally above private market repo rates that prevail in normal times, intending for the facility to be used in periods of market stress but not as markets recover (Arnold 2023d; Fed 2020e). The former head of the FRBNY's Markets Group, Daleep Singh, explained: We made sure the pricing of these facilities was set at a backstop rate, such that they would self-liquidate when market conditions normalized, and also to ensure that we weren't creating adverse pricing for domestic actors relative to international institutions (Singh 2023).

Finally, in some cases, the central bank converted a temporary swap arrangement into a standing arrangement rather than terminate it. Standing swap lines are valuable because they remain available for reactivation or targeted use on short notice (see Key Design Decision No. 1(B), Purpose and Type). One case in point is that during the banking stresses of March 2023, the six central banks of the Major Bank Swap Network again took coordinated action to improve the provision of US dollar liquidity. Those banks that had weekly downstream dollar auctions increased them to daily operations, and those without operations implemented them so the weekly auctions began on the same day (BBC News 2023).

[28] The Fed's temporary swap lines with the BoC, BoE, BOJ, ECB, and SNB were deployed in 2010 during the Sovereign Debt Crisis and converted into standing swap lines in October 2013, forming the Major Bank Swap Network.

Conclusion

Since the GFC, currency swap lines and repos have become a key policy tool. Central banks used them very successfully during the GFC, SDC, COVID-19 pandemic, and other crises. Most prominently, the Fed lent more than USD 1 trillion to other central banks through its swap lines during the GFC and COVID-19 crises combined (Fed n.d.a; Perks et al. 2021).

Since 2013, six lending central banks of advanced economies (BoC, BoE, BOJ, ECB, Fed, and SNB) have maintained a standing network—the Major Bank Swap Network—which includes major reserve currencies and all G-7 countries. The network allows any participating central bank to access uncapped amounts of any other participant's currency in exchange for its own. Mostly, central banks have used this network to access dollars from the Fed during the COVID-19 crisis and, to a lesser extent, in March 2023. They then coordinated simultaneous downstream US dollar auctions for domestic financial institutions at fixed interest rates and full allotment (Fed 2020a; Fed 2023b).

Outside such standing arrangements, lending central banks continue to establish or activate other swap lines as needed to expand access to their currencies outside their borders. Although Lender banks tend to be conservative about whom they choose as counterparties, they often act as regional lenders of last resort, partnering with countries with which they have close economic, trade, financial, or cultural ties. We saw this during the GFC with the three Scandinavian central banks' swap lines to Iceland and the RBI's swap lines with Bhutan, the Maldives, and Sri Lanka, under the SAARC swap framework.

Lending central banks have increasingly used repo lines rather than currency swaps to limit their exposure to currency risk. The ECB has used bilateral repo lines since the GFC. In 2020, the Fed and ECB adopted broad standing repo frameworks to extend liquidity in their currencies to countries with whom they have not established swap lines. The Fed's FIMA Repo Facility permits access to virtually any central bank; the ECB's EUREP framework is available to a large but unspecified number of central banks. In both facilities, central banks that have sufficient dollar- or euro-denominated collateral may exchange it for currency once they have applied for and received access.

The PBOC has also created a major swap network, entering into arrangements with 40 countries since 2008 (Horn et al. 2023; PBOC 2022). Although originally instituted to promote renminbi trade and settlement, these lines are increasingly being used for liquidity and financial stability. The PBOC has been more willing to enter into swaps with emerging market economies than have the major advanced economy central banks. Thus, this developing resource reaches many countries that do not have swap arrangements with Lender banks of other major reserve currency countries, and scholars are beginning to view the PBOC as a major LOLR force (Horn et al. 2023).

The regional swap facilities, such as the ASA, the CMIM, and the SAARC, were also formed to provide liquidity and balance of payments support to smaller and emerging market economies without access to swap arrangements with reserve currency Lender banks. To date, these facilities have not played a large role as LOLR resources, but there is potential for them to evolve and grow their influence in the future.

In their typical form, swaps are simple agreements that embody minimal exchange or credit risk. Many standard terms (exchange rate, interest rate, length) are set forth in the framing agreement, so draws require limited administration. The counterparties may customize their agreements to further protect the Lender's balance sheet, exert influence over the use of drawn funds, or ensure long-term stabilizing efforts are achieved by Borrower countries. Such customization tends to occur in swaps between an advanced-economy Lender country and an emerging market Borrower country. Swaps may be just one component of a broader program that the Lender country, other countries, or the IMF provides to the Borrower. However, overall, we have observed only limited customization in these cases, perhaps because of the increased use of repos.

The use of central bank swaps and repos as liquidity tools is likely to continue. The greater range of swap and repo liquidity facilities creates new opportunities for all types of central banks to access the currencies they need in a crisis.

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Appendix

Figure 13: Frequently Cited Central Bank Names and Abbreviations

Country/Region	Central Bank	Abbreviation
Argentina	Central Bank of Argentina	BCRA
Brazil	Banco Central do Brasil	BCB
Canada	Bank of Canada	BoC
China	People's Bank of China	PBOC
Denmark	Danmarks Nationalbank	DNB
Euro area	European Central Bank	ECB
Hong Kong	Hong Kong Monetary Authority	HKMA
Hungary	Magyar Nemzeti Bank	MNB
Iceland	Central Bank of Iceland	CBI
India	Reserve Bank of India	RBI
Ireland	Central Bank of Ireland	CBIR
Japan	Bank of Japan	BOJ
Korea	Bank of Korea	BOK
Mexico	Bank of Mexico	Banxico
Mongolia	Bank of Mongolia (aka Mongolbank)	BOM
New Zealand	Reserve Bank of New Zealand	RBNZ
Poland	National Bank of Poland	NBP
Singapore	Monetary Authority of Singapore	MAS
Sweden	Sveriges Riksbank	Riksbank
Switzerland	Swiss National Bank	SNB
United Kingdom	Bank of England	BoE
United States	Federal Reserve	Fed

Source: Authors' compilation.